

Article

Scheduling Cluster Tools with Multi-Space Process Modules and a Multi-Finger-Arm Robot in Wafer Fabrication Subject to Wafer Residency Time Constraints

Lei Gu , Naiqi Wu *, Yan Qiao , Siwei Zhang and Tan Li

Institute of Systems Engineering and Department of Engineering Science, Faculty of Innovation Engineering, Macau University of Science and Technology, Taipa, Macao 999078, China; 2109853vmi30001@student.must.edu.mo (L.G.); yqiao@must.edu.mo (Y.Q.); swzhang@must.edu.mo (S.Z.); 1909853omi30001@student.must.edu.mo (T.L.)

* Correspondence: nqw@must.edu.mo; Tel.: +853-88971991

Abstract: To increase productivity, more sophisticated cluster tools are developed. To achieve this, one of the ways is to increase the number of spaces in a process module (PM) and the number of fingers on a robot arm as well, leading to a cluster tool with multi-space PMs and a multi-finger-arm robot. This paper discusses the scheduling problem of cluster tools with four-space PMs and a four-finger-arm robot, a typical tool with multi-space PMs and a multi-finger-arm robot adopted in modern fabs. With two arms in such a tool, one is used as a clean one, while the other is used as a dirty one. In this way, wafer quality can be improved. However, scheduling such cluster tools to ensure the residency time constraints is very challenging, and there is no research report on this issue. This article conducts an in-depth analysis of the steady-state scheduling for this type of cluster tools to explore the effect of different scheduling strategies. Based on the properties, four robot task sequences are presented as scheduling strategies. With them, four linear programming models are developed to optimize the cycle time of the system and find feasible schedules. The performance of these strategies is dependent on the activity parameters. Experiments are carried out to test the effect of different parameters on the performance of different strategies. It shows that, given a group of parameters, one can apply all the strategies and choose the best result obtained by one of the strategies.

Keywords: semiconductor manufacturing; cluster tools; scheduling; linear program



Citation: Gu, L.; Wu, N.; Qiao, Y.; Zhang, S.; Li, T. Scheduling Cluster Tools with Multi-Space Process Modules and a Multi-Finger-Arm Robot in Wafer Fabrication Subject to Wafer Residency Time Constraints. *Appl. Sci.* **2024**, *14*, 9490. <https://doi.org/10.3390/app14209490>

Academic Editor: Gerard Ghibaudo

Received: 25 August 2024

Revised: 11 October 2024

Accepted: 15 October 2024

Published: 17 October 2024



Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Cluster tools have been demonstrated to be adaptable, reconfigurable, and efficient for wafer fabrication in semiconductor manufacturing [1,2], resulting in increased productivity [3], reduced cycle time [3–5], high space utilization [2,4], and decreased capital costs. As illustrated in Figure 1, cluster tools are generally composed of several process modules (PMs), a wafer-handling robot, and two loadlocks (LLs) for loading and unloading wafer cassettes, usually holding 25 wafers with the same recipe [6,7]. Wafer processing in PMs adopts single-wafer processing technology, such that each PM can process only one wafer at a time. A raw wafer in a cassette is delivered to PMs for processing by the robot in a pre-specified order and finally returned to the cassette after all operations are completed.

To increase the productivity, more sophisticated cluster tools are developed. One of the methods of cluster-tool development is to increase the number of spaces in a PM and the number of fingers on a robot arm to form a cluster tool with multi-space PMs and a multi-finger-arm robot. As shown in Figure 2, the cluster tools discussed in this article are a typical type of cluster tool adopted in modern fabs and are different from the ones with single-space PMs illustrated in Figure 1. The discussed cluster tool shown in Figure 2 consists of several four-space PMs, a dual-arm–dual-finger robot, and several LLs. As shown in Figure 2, in a four-space PM, at a time, two spaces face the door, such that the

robot can load wafers into them or unload wafers from them, while the other two spaces are located at the opposite position. The internal chamber can then rotate to exchange the positions taken by the spaces so that the robot can access the other two spaces. In this way, the robot can access all the spaces in the PMs according to a certain order. For the dual-arm–dual-finger robot, each arm has two fingers. Although both fingers in an arm can hold a wafer, they cannot work independently. Also, the two arms are structurally tightly coupled so that only one of them can load/unload at a time. Thus, when one of the robot arms performs an unloading or loading task at a PM or an LL, the two fingers on it should enter the PM or LL together to pick up or place two wafers (or one wafer if there is only one wafer in the spaces at the door of the PM or there is only one wafer held by one of the fingers). After the robot loads four wafers into a PM, a rotation operation of the internal chamber of the PM is required before starting the processing so that the two wafers that are loaded into the PM earlier can be unloaded before the other two upon their completion. This first-in-first-out rule minimizes the difference in sojourn time between the wafers loaded earlier and those loaded later. Moreover, there are several LLs (normally four) to make sure that raw wafers can enter the tool for processing in time. Thus, LLs are not the bottleneck such that the throughput of a tool is not determined by them. Also, this implies that once the robot comes to an LL to perform an unloading/loading task, it can do so immediately without waiting.

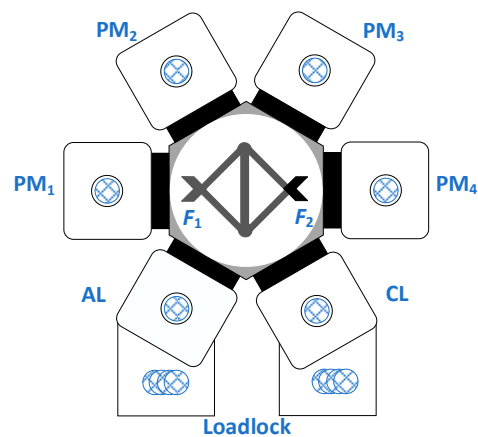


Figure 1. A cluster tool with single-space PMs.

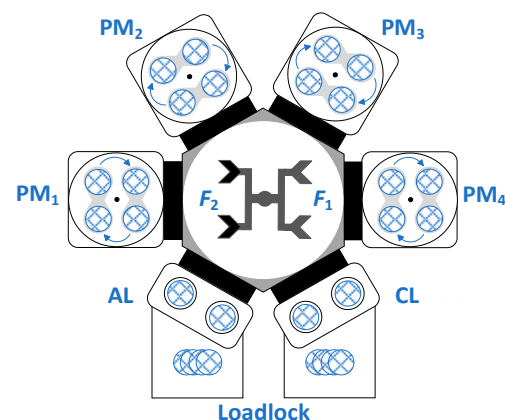


Figure 2. A cluster tool with four-space PMs.

As the wafer circuit shrinks down to several nanometers, in wafer fabrication, contamination should be avoided by meeting various requirements. One of such requirements is the strict wafer residency time constraints (RTCs) [8,9]. With RTCs, once a wafer is processed in a PM, it should be removed from the PM within a limited time; otherwise, high temperature and chemical gas would contaminate or damage the wafer in it. Another

requirement is the right use of fingers for wafer grasping. To realize this requirement, robot fingers are classified into two types: clean and dirty ones. The fingers dedicated to grasping raw wafers (called dirty wafers as well) are called dirty fingers, while fingers employed to grasp processed wafers are called clean fingers. By doing so, cross-contamination can be avoided. In real-life semiconductor fab applications, for cluster tools shown in Figure 2, two fingers on one of the robot arms are used as clean ones, while the other two fingers on the other robot arm are treated as dirty ones. The cluster tools shown in Figure 2 are normally used for chemical vapor deposition processes.

The cluster tool illustrated in Figure 1 is a cluster tool with single-space PM and single-finger-arm robot (S^3F -CT), while the cluster tool illustrated in Figure 2 is a cluster tool with four-space PM and four-finger-arm robot (SF^3 -CT). Previous studies provide the optimal scheduling strategy for S^3F -CT. However, existing studies do not deal with the practical situations that a robot arm is constrained to dedicate a part of robot activities to as stated in this paper. To the best of the authors' knowledge, there is no research report on the scheduling of SF^3 -CTs, as the existing studies related to cluster tool scheduling mainly focus on S^3F -CTs.

Two or more LLs in a cluster tool can ensure that there are enough raw wafers entering the tool for processing, such that the tool can operate under a steady state most of the time. Practically, the wafer-processing time in a PM is quite longer than the robot task time [10,11]. Nevertheless, the robot task time in a cycle might be longer than the wafer completion time at a step. Under a steady state, if the robot task time in a cycle is longer than the wafer completion time at the bottleneck step, the cluster tool is said to be transport-bound. In this situation, the robot is always busy. If the robot task time in a cycle is shorter than the wafer completion time at the bottleneck step, then the cluster tool is said to be process-bound. In this situation, the robot has idle time. For process-bound cluster tools with a single-arm robot, a backward strategy is optimal [11], while for process-bound cluster tools with a dual-arm robot, a swap strategy is efficient [12]. Significant advancements have been made in the area of modeling, performance assessment, control, and scheduling of cluster tools under the steady state [7,13–18]. However, RTCs are not considered in those studies.

With RTCs, Kim et al. [19], Lee and Park [20], and Rostami et al. [8] propose several methods to find an optimal and feasible cyclic schedule for cluster tools with a dual-arm robot. Furthermore, Wu et al. [6,21] conduct studies on scheduling cluster tools equipped with a single-arm robot/dual-arm robot from a perspective of control theory so as to obtain an optimal and feasible schedule efficiently. Lim et al. [22] introduce a novel category of robot task sequences that are more effective in meeting RTCs for situations with unbalanced processing parameters among steps. Ko et al. [23] propose an approach for cluster tools with a dual-arm robot handling multiple wafer types, aiming to satisfy RTCs and achieve the minimum cycle time by identifying an optimal release sequence for different wafer types. For cluster tools with processing-time variation and RTCs, Lim et al. [24] suggest a flexible strategy to improve the schedule robustness. For the scheduling problem of multi-cluster tools with RTCs, Zhu et al. [25] find a feasible steady-state schedule in dealing with two wafer types concurrently. For re-entrant processes with RTCs, efforts have also been made in the modeling, control, and scheduling of cluster tools with single-arm and dual-arm robots [26,27]. Meanwhile, the studies in [28,29] examine how activity time variation affects the wafer delay in PMs for cluster tools with single-arm and dual-arm robots. Yu et al. [30–32] suggest partially loading wafers in parallel PMs to improve the productivity of cluster tools. However, they do not consider multi-space PMs, nor address the issue of cross-contamination between dirty and clean wafers.

For S^3F -CTs, only after a wafer enters a PM can the PM start its processing activity, implying that the processing activities follow the robot tasks. As a result, it is important to carefully determine the robot task sequence [6,21]. However, for the tools addressed in this work, except for the processing activities, the internal chambers of PMs should perform rotation operations such that the robot can correctly load wafers into PMs for processing. Notice that if a PM is still performing a rotation operation when the robot arrives at the PM

for loading or unloading, the robot has to wait. This implies that rotation operations might delay the robot task time, too. Therefore, for SF³–CTs shown in Figure 2, it is necessary to analyze how the rotation operations in PMs affect the robot task schedule. This makes the obtained results for S³F–CTs in the existing studies not directly applicable to SF³–CTs. Furthermore, RTCs and the requirement for the right use of clean and dirty fingers make the scheduling problem of SF³–CTs much more complicated.

Note that, for the addressed tool, four wafers can be processed concurrently in a PM. However, each time, only two wafers can be loaded/unloaded, resulting in the increase in time delay of these wafers in a PM. Thus, how to reduce such a time delay is an important issue. Up to now, there is no research report on this topic, while practical applications require corresponding techniques to provide solution methods for that, i.e., there is a gap between the existing research achievements and the practical application demand. This motivates us to conduct this study. It should be pointed out that, as discussed above, although there are extensive studies on scheduling cluster tools with RTCs, the time delay caused by loading wafers into a PM more than once time makes the RTCs much more difficult to satisfy. Furthermore, due to the fact that the robot arm is restricted from performing certain tasks, the widely used scheduling strategies are not applicable, since the existing methods assume that there is no restriction on the robot arms. In summary, for the addressed problem, we face new challenges. With the new challenges, this work aims to examine new scheduling strategies and explore the effect of different scheduling strategies on obtaining feasible schedules. To do so, based on the properties of the system, four scheduling strategies are presented, and methods are developed to find feasible and optimal schedules. Then, experiments are conducted to test the performance of the methods on the capability for finding feasible and optimal schedules.

The rest of this article is organized as follows. In the next section, a detailed description of the problem is given. Then, the properties of the system are analyzed in Section 3. In Section 4, linear programs are established to find optimal and feasible schedules based on the presented robot task sequences. In Section 5, comparison experiments are carried out to show the application of the proposed method. Section 6 concludes this work.

2. The Processes and Scheduling Strategies

2.1. The Operation Processes

In semiconductor fabs, wafers are associated with recipes, specifying the wafer flow patterns (WFPs) and processing parameters. WFPs specify the processing routes of wafers. For an SF³–CT, WFP is defined as $(m_1, m_2, m_3, \dots, m_n)$, with n being the number of process steps required to process wafers and m_i the number of parallel PMs at Step i . LLs are treated as Steps 0 and $n+1$ for holding raw and completed wafers, respectively. The robot arm with two dirty fingers is denoted by F_1 , and we call this arm the dirty arm, while the other robot arm with two clean fingers is represented by F_2 , and we call it the clean arm. Two raw wafers are unloaded from an LL by F_1 and delivered to a PM at Step 1 for processing. Once they are processed at Step 1, they are treated as clean wafers. Then, F_2 should deliver them to Steps 2, $\dots$, n for processing sequentially. After all operations are completed, they are returned to LL by F_2 .

To schedule a cluster tool, determining the robot task sequence is vitally important so as to maximize the tool's productivity and meet RTCs. The robot tasks include unloading, loading, and moving; U_i and L_i are used to denote the robot tasks of unloading and loading at a PM of Step i , $i \in \mathbb{N}_n = \{0, 1, \dots, n\}$. Since the time taken for a robot task is quite short in practice, the time taken to perform U_i , $i \in \mathbb{N}_n \setminus \{0\}$, and L_i , $i \in \mathbb{N}_n$, is the same and denoted by β . Note that before the robot unloads two clean wafers from an LL, the alignment of the wafers should be carried out first, resulting in the time to perform U_0 being relatively longer. Thus, β_0 is used to denote the time taken for U_0 . The moving task from a PM at Step i (or an LL at Step 0) to a PM at Step j (or an LL at Step 0) is denoted by $M_{i,j}$, $i, j \in \mathbb{N}_n$, and the time taken for $M_{i,j}$ is denoted by μ . Note that for $M_{i,j}$, $i, j \in \mathbb{N}_n$, if $i = j$, such a robot-moving task is called a rotation task. Assume that a robot arm is in the front of

a PM, if the robot performs a rotation task, then the other robot arm should rotate to the front of the PM after the rotation. Furthermore, a swap operation of the robot includes an unloading task, a rotation task, and a loading task.

At the idle state, all four spaces in a PM are empty. Then, after two wafers are loaded into two of the spaces, the internal chamber of the PM rotates to make the other two empty spaces face the door, such that the robot can load another two wafers into the PM. Similarly, a rotation operation should be performed when the four wafers are processed and should be removed from the PM by the robot consecutively. Symbol v is used to denote the time taken for a rotation operation of a PM. The wafer-processing time in a PM at Step i is denoted by α_i , $i \in \mathbf{N}_n$. Assume that there are four wafers in a PM at Step i . Let $W_{i,12}$ and $W_{i,34}$ be the four wafers that are to be processed in a PM at Step i concurrently and $W_{i,12}$ be the two wafers that are loaded into the PM earlier than $W_{i,34}$. Notice that after $W_{i,12}$ are loaded into the PM at Step i , the internal chamber rotates such that $W_{i,34}$ can be loaded into the PM. Then, with $W_{i,34}$ being loaded, the rotation operation should be executed again to move the two spaces holding $W_{i,12}$ to the door position such that the PM can start to process wafers. After the processing activity is completed, $W_{i,12}$ can be removed from the PM earlier than $W_{i,34}$. In this way, one can ensure that the wafers loaded into the PM earlier can be removed from the PM earlier.

Note that, under a steady state, there are four wafers being processed in a PM at Step i . When the robot comes to the PM for unloading $W_{i,12}$, the processing activity at the PM may not be completed. Thus, the robot has to wait here for some time, and such a robot-waiting task is denoted by A_{i1} . After $W_{i,12}$ is removed from the PM, the internal chamber should rotate such that the spaces holding $W_{i,34}$ rotate to the door position of the PM. Also, when the robot comes to the PM again for unloading $W_{i,34}$, the rotation operation of the PM may not be completed. At this time, the robot should wait, and the robot-waiting activity is denoted by A_{i2} . Furthermore, when the robot loads $W_{i,12}$ into a PM at Step i , it implies that the PM is empty. Thus, the robot can load $W_{i,12}$ immediately without waiting. After $W_{i,12}$ are loaded into the PM, the PM should rotate such that other two empty spaces rotate to the door position of the PM. When the robot comes to the PM again and is ready to load $W_{i,34}$ into the PM, the internal chamber of the PM may not complete the rotation operation. This means that the robot should wait at the PM. Such a robot-waiting activity is denoted by A_{i3} . In addition, the times for A_{i1} , A_{i2} , and A_{i3} are represented by ω_{i1} , ω_{i2} , and ω_{i3} , respectively.

To meet RTCs, after the wafers are processed in a PM, they should leave the PM within a limited time window. Let δ_i denote such a limited time at Step i , $i \in \mathbf{N}_n \setminus \{0\}$, and d_i denote the wafer delay time of a wafer in a PM at Step i , i.e., the time duration starting from the time point at which the processing activity is completed to the time point at which the robot grasps the wafer for unloading. We use $\tau_{i,12}$ and $\tau_{i,34}$ to denote the wafer sojourn time of $W_{i,12}$ and $W_{i,34}$, respectively. Then, if a schedule for an SF³-CT is feasible, the following constraints should be met:

$$\alpha_i \leq \tau_{i,12} \leq \alpha_i + \delta_i, i \in \mathbf{N}_n \setminus \{0\} \quad (1)$$

$$\alpha_i \leq \tau_{i,34} \leq \alpha_i + \delta_i, i \in \mathbf{N}_n \setminus \{0\} \quad (2)$$

2.2. Robot Task Sequences

Assume that there are n (≥ 2) steps for an SF³-CT. To maximize the productivity of an SF³-CT, under the steady state, each PM should be full of wafers. To maximize the productivity and ensure the feasibility (i.e., meet RTCs) in operating an SF³-CT, it is vital to properly determine a robot task sequence that can be performed periodically. In [11,12], the authors prove that backward and swap are the optimal scheduling strategies for single-arm and dual-arm cluster tools, respectively. With clean and dirty robot fingers, in terms of operation strategies, an SF³-CT is neither a single-arm cluster tool nor a dual-arm cluster tool but a hybrid one that is equivalent to a combination of them; at some steps, swap operations can be applied, while they are not applicable at some other steps. Due to this

property, there are two types of operation strategies for an SF³–CT, one is the backward strategy and the other is hybrid strategy, which combines the backward and swap strategies. Next, we present four possible robot task sequences as the operation strategies. Note that the following robot task sequences are presented based on the assumption that the tool operates under a steady state with each PM full of wafers and the clean robot arm F_2 is at a PM of Step n .

The first robot task sequence is a backward strategy, called one-time backward sequence (OBS). For OBS, the clean robot arm F_2 first performs task sequence $\sigma_1 = \langle U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-1} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-2} \rightarrow \dots \rightarrow M_{3,1} \rightarrow U_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,1} \rightarrow U_1 \rightarrow M_{1,2} \rightarrow L_2 \rangle$ to handle clean wafers. Then, the dirty robot arm F_1 performs robot task sequence $\sigma_2 = \langle M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rangle$ to load raw wafers to the system. Note that, by $M_{1,n}$ in σ_2 , a state is reached such that F_2 is at a PM of Step n . Thus, by sequentially performing σ_1 and σ_2 , a production cycle is finished, with four wafers being processed. Notice that, by this strategy, after σ_1 is performed, a PM at Step 1 is emptied.

The second robot task sequence is a hybrid strategy called one-time hybrid task sequence (OHTS). For OHTS, the clean arm F_2 first performs a robot task sequence $\sigma_3 = \langle U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-1} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-2} \rightarrow \dots \rightarrow M_{4,2} \rightarrow U_2 \rightarrow M_{2,3} \rightarrow L_3 \rightarrow M_{3,2} \rightarrow U_2 \rightarrow M_{2,3} \rightarrow L_3 \rangle$ to handle clean wafers. Then, arms F_1 and F_2 collaboratively perform robot task sequence $\sigma_4 = \langle M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow \text{a swap operation at Step 1} \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow \text{a swap operation at Step 1} \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,n} \rangle$ to load raw wafers into the system. Note that, by σ_4 , the system reaches a state such that F_2 is at a PM of Step n . Thus, by sequentially performing σ_3 and σ_4 , a production cycle is finished with four wafers being processed. Different from OBS, after performing σ_3 , a PM at Step 2 is emptied, but no PM at Step 1 is emptied.

The third robot task sequence is also a backward strategy, called two-time backward sequence (TBS). For TBS, F_2 first performs robot task sequence $\sigma_5 = \langle U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-2} \rightarrow \dots \rightarrow M_{3,1} \rightarrow U_1 \rightarrow M_{1,2} \rightarrow L_2 \rangle$ to handle clean wafers. Then, $\sigma_6 = \langle M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rangle$ is performed by F_1 and F_2 together to load raw wafers into the system. After that, σ_5 and σ_6 are sequentially performed again. At this time, each PM in the tool is full of wafers again. Therefore, a production cycle is conducted by performing $\sigma_5 \rightarrow \sigma_6 \rightarrow \sigma_5 \rightarrow \sigma_6$, such that each step completes four wafers in such a cycle. By TBS, after performing σ_5 for Steps 1– n , two processed wafers in a PM are moved to the next step for further processing, with two spaces of a PM at Step 1 being emptied.

The last robot task sequence is also a hybrid strategy, called two-time hybrid task sequence (THTS). For THTS, F_2 first performs robot task sequence $\sigma_7 = \langle U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-2} \rightarrow \dots \rightarrow M_{4,2} \rightarrow U_2 \rightarrow M_{2,3} \rightarrow L_3 \rangle$ to handle clean wafers. After performing σ_7 , $\sigma_8 = \langle M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow \text{a swap operation at Step 1} \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,n} \rangle$ is performed by F_1 and F_2 collaboratively to load raw wafers into the system. Then, σ_7 and σ_8 are sequentially performed again. At this time, each PM in the tool is full of wafers again. Therefore, similar to TBS, a production cycle is conducted by sequentially performing $\sigma_7 \rightarrow \sigma_8 \rightarrow \sigma_7 \rightarrow \sigma_8$, such that each step completes four wafers in such a cycle. This strategy is similar to TBS; the difference is that, after performing σ_7 , no processed wafer at Step 1 is moved to Step 2, and two spaces of a PM at Step 2 are emptied.

With these four robot task sequences, the clean and dirty fingers are correctly used. In σ_1 – σ_8 , the robot-waiting time is not considered. Although a robot task sequence in each of σ_1 – σ_8 is determined, it does not present a schedule since the start and end times of the activities are not given. However, as discussed above, if the robot-waiting time is decided, a schedule is obtained. Thus, to find a feasible and optimal schedule under one of the four robot task sequences, we need to intend to increase the robot waiting time taken by A_{i1} . Thus, the key to scheduling an SF³–CT is to determine the robot-waiting time, such that the

RTCs are met and the productivity of the system is maximized. The next Section aims to do so.

In this Section, the operational processes of semiconductor fabs using cluster tools are discussed with a focus on wafer flow patterns and the robot task sequence. It describes how dirty and clean wafers are treated separately for a multi-arm and multi-finger robot and how RTC should be met to avoid contamination. It outlines the importance of properly sequencing robot tasks to ensure productivity while maintaining RTCs and provides the foundational details on how wafers are processed in parallel within multi-space process modules.

3. Scheduling Analysis

In this section, we describe the time aspect under OBS, OHTS, TBS, and THTS, respectively. Then, four linear programs are established to find optimal and feasible schedules for an SF³-CT.

3.1. Temporal Properties Under OBS

Let θ_i be the time for processing four wafers at Step i . Under the steady state, to complete four wafers by a PM at Step n , by OBS, with the robot-waiting activities being considered, task sequence $\sigma_9 = \langle U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{i+1} \rightarrow M_{n+1,n-1} \rightarrow A_{(n-1)1} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-1} \rightarrow A_{(n-1)2} \rightarrow U_{n-1} \rightarrow M_{n-1,i} \rightarrow A_{n3} \rightarrow L_n \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rangle$ should be performed.

From Figure 3, we can see the task sequence of the robot between PM₁ and PM₂ and between PM₁ and LL as well. In the figure, the red arrows present the wafer movement direction; blue arrows indicate the next step in the process; blank circles represent empty LLs; purple/blue circles represent wafers. In the steps with a yellow circular background, the internal chamber of the PM rotates just before starting to process wafers. Note that different colored wafers are used to clarify the robot task sequence and do not represent different wafers. With m_n parallel PMs serving for Step n , according to σ_9 , the time for processing four wafers at the step can be calculated using Equation (3):

$$\theta_n = (\alpha_n + d_n + 8\beta + 7\mu + v + \omega_{n2} + \omega_{n3} + \omega_{(n-1)1} + \omega_{(n-1)2})/m_n \quad (3)$$

Furthermore, to complete four wafers by a PM of Step i , $i \in \{3, \dots, n-1\}$ and $n > 2$, by OBS, with the robot-waiting activities being considered, task sequence $\sigma_{10} = \langle U_i \rightarrow M_{i,i+1} \rightarrow L_{i+1} \rightarrow M_{i+1,i} \rightarrow A_{i2} \rightarrow U_i \rightarrow M_{i,i+1} \rightarrow A_{(i+1)3} \rightarrow L_{i+1} \rightarrow M_{i+1,i-1} \rightarrow A_{(i-1)1} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rightarrow L_i \rightarrow M_{i,i-1} \rightarrow A_{(i-1)2} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rightarrow A_{i3} \rightarrow L_i \rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rangle$ should be performed. With m_i parallel PMs serving for Step i , according to σ_{10} , the time for processing four wafers at the step can be calculated using Equation (4):

$$\theta_i = (\alpha_i + d_i + 8\beta + 7\mu + v + \omega_{(i+1)3} + \omega_{i2} + \omega_{i3} + \omega_{(i-1)1} + \omega_{(i-1)2})/m_i, i \in \{3, \dots, n-1\} \text{ and } n > 2 \quad (4)$$

Similarly, to complete four wafers by a PM of Step 1, by OBS, with the robot-waiting activities being considered, task sequence $\sigma_{11} = \langle U_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow M_{1,2} \rightarrow A_{23} \rightarrow L_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{13} \rightarrow L_1 \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rangle$ should be performed. With m_1 parallel PMs serving for Step 1, according to σ_{11} , the time for processing four wafers at the step can be calculated with Equation (5):

$$\theta_1 = (\alpha_1 + d_1 + 2\beta_0 + 6\beta + 7\mu + v + \omega_{23} + \omega_{12} + \omega_{13})/m_1 \quad (5)$$

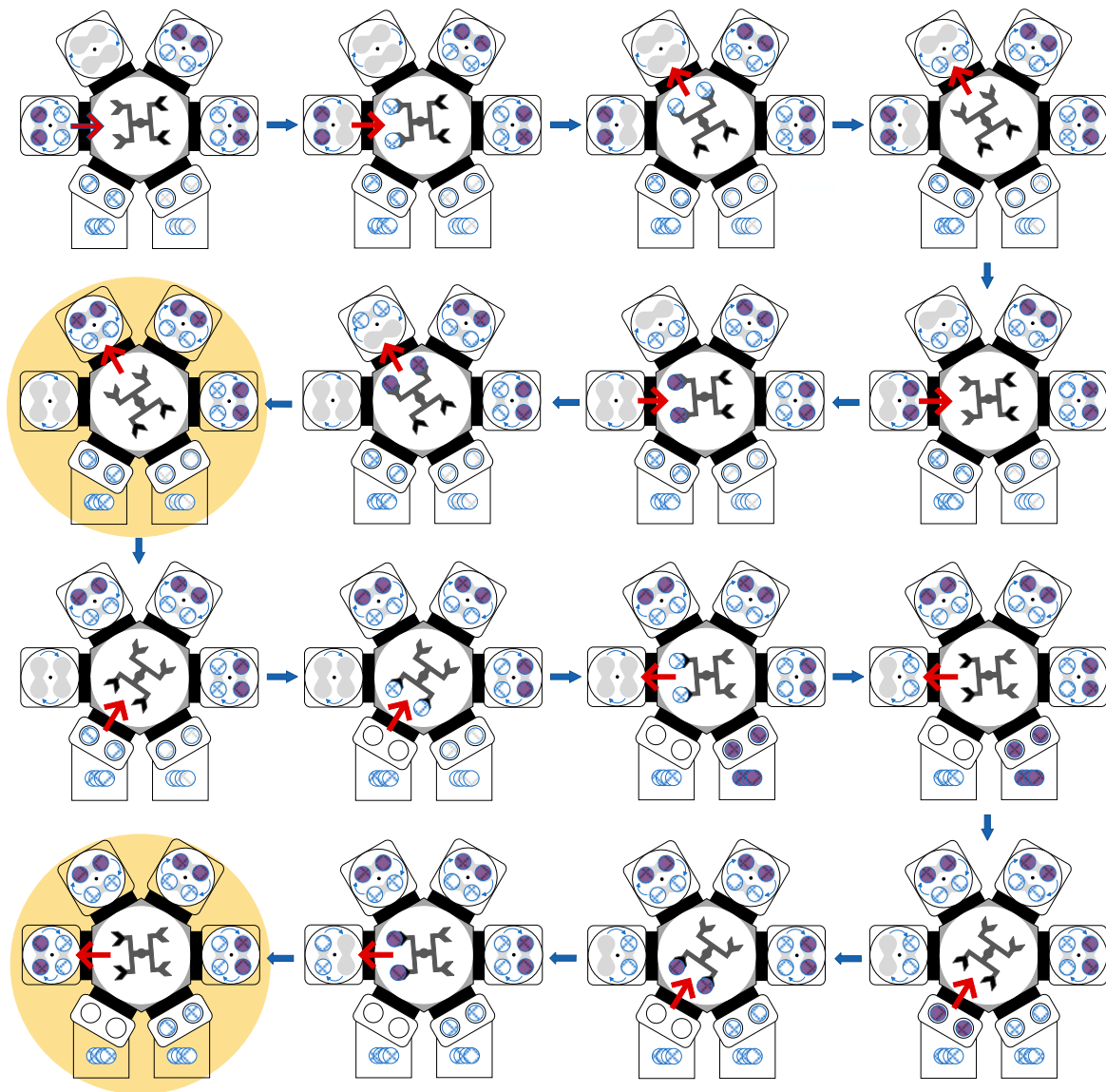


Figure 3. Description of robot movements under the OBS strategy.

In σ_9, σ_{10} , and σ_{11} , $A_{i2}, i \in \mathbb{N}_n \setminus \{0\}$ is caused by the rotation of the internal chamber of a PM for Step i . Notice that after the robot performs an unloading task at the PM in σ_9, σ_{10} , or σ_{11} for the first time, the internal chamber of the PM starts to rotate. At the same time, the robot performs task sequence $\langle M_{i,i+1} \rightarrow L_{i+1} \rightarrow M_{i+1,i} \rangle$. Then, A_{i2} should be performed, and the time taken for A_{i2} is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{i,i+1} \rightarrow L_{i+1} \rightarrow M_{i+1,i} \rangle$. Thus, we have the following Equation:

$$\omega_{i2} = \max(v - (2\mu + \beta), 0), i \in \mathbb{N}_n \setminus \{0\} \quad (6)$$

In σ_9, σ_{10} , and σ_{11} , $A_{i3}, i \in \mathbb{N}_n \setminus \mathbb{N}_1$ is caused by the rotation of the internal chamber of a PM for Step i as well. Notice that after the robot performs a loading task at Step i in σ_9, σ_{10} , and σ_{11} for the first time, the internal chamber of the PM starts to rotate. At the same time, the robot performs task sequence $\langle M_{i,i-1} \rightarrow A_{(i-1)2} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rangle$. Then, A_{i3} should be performed, and the time taken for A_{i3} is determined by the longest one between the time taken for a rotation operation of a PM for Step i and $\langle M_{i,i-1} \rightarrow A_{(i-1)2} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rangle$. Thus, we have the following Equation:

$$\omega_{i3} = \max(v - (\omega_{(i-1)2} + 2\mu + \beta), 0), i \in \mathbb{N}_n \setminus \mathbb{N}_1 \quad (7)$$

Similarly, the time taken for A_{13} is determined by the longest one between the time taken for a rotation operation of a PM for Step 1 and $\langle M_{1,0} \rightarrow U_0 \rightarrow M_{0,1} \rangle$, and thus, we have the following Equation:

$$\omega_{13} = \max(v - (2\mu + \beta_0), 0) \quad (8)$$

According to σ_9 and σ_{10} , the time taken for task sequence $\langle M_{i,i-1} \rightarrow A_{(i-1)2} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rightarrow A_{i3} \rightarrow L_i \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rangle$ equals the wafer sojourn time of $W_{i,12}$, $i \in \mathbf{N}_n \setminus \mathbf{N}_1$. Furthermore, the time taken for task sequence $\langle$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rightarrow U_i \rightarrow M_{i,i+1} \rightarrow L_{i+1} \rightarrow M_{i+1,i} \rightarrow A_{i2} \rangle$ equals the wafer sojourn time of $W_{i,34}$, $i \in \mathbf{N}_n \setminus \mathbf{N}_1$. Then, we have the following Equation:

$$\tau_{i,12} = \alpha_i + d_i + v + 2\beta + 2\mu + \omega_{(i-1)2} + \omega_{i3}, i \in \mathbf{N}_n \setminus \mathbf{N}_1 \quad (9)$$

$$\tau_{i,34} = \alpha_i + d_i + v + 2\beta + 2\mu + \omega_{i2}, i \in \mathbf{N}_n \setminus \mathbf{N}_1 \quad (10)$$

According to σ_{11} , the time taken for task sequence $\langle M_{1,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{13} \rightarrow L_1 \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rangle$ equals the wafer sojourn time of $W_{1,12}$. Furthermore, the time taken for task sequence $\langle$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rightarrow U_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,1} \rightarrow A_{12} \rangle$ equals the wafer sojourn time of $W_{1,34}$. Then, we have the following Equation:

$$\tau_{1,12} = \alpha_1 + d_1 + v + \beta + \beta_0 + 2\mu + \omega_{13} \quad (11)$$

$$\tau_{1,34} = \alpha_1 + d_1 + v + 2\beta + 2\mu + \omega_{12} \quad (12)$$

For OBS, with the robot-waiting activities being considered, σ_1 and σ_2 can be rewritten as $\langle U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow A_{(n-1)1} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-1} \rightarrow A_{(n-1)2} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow A_{n3} \rightarrow L_n \rightarrow M_{n,n-2} \rightarrow \dots \rightarrow M_{3,1} \rightarrow A_{11} \rightarrow U_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow M_{1,2} \rightarrow A_{23} \rightarrow L_2 \rangle$ and $\langle M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{13} \rightarrow L_1 \rightarrow M_{1,n} \rightarrow A_{n1} \rangle$, respectively. Then, under OBS, the robot cycle time ψ_1 formed by σ_1 and σ_2 together is given as follows:

$$\psi_1 = 4(n+1)\mu + 2(2n+1)\beta + 2\beta_0 + \sum_1^n \omega_{i1} + \sum_1^n \omega_{i2} + \sum_1^n \omega_{i3} \quad (13)$$

3.2. Temporal Properties Under OHTS

By OHTS, with the robot-waiting activities being considered, similar to OBS, task sequence σ_9 or σ_{10} is executed so as to complete four wafers by a PM for Step i , $i \in \mathbf{N}_n \setminus \mathbf{N}_2$. Thus, with m_i parallel PMs serving for Step i , according to σ_9 and σ_{10} , the time for processing four wafers at Step i , $i \in \mathbf{N}_n \setminus \mathbf{N}_2$, can be calculated using Equations (3) and (4). We can see the task sequence of the robot among PM₁, PM₂, and LL as presented in Figure 4, where the meaning of colored arrows is same as that in Figure 3. In the steps with a yellow circular background, the internal chamber of the PM rotates just before starting to process wafers.

For Step 2, $n > 2$, to complete four wafers at a PM, task sequence $\sigma_{12} = \langle U_2 \rightarrow M_{2,3} \rightarrow L_3 \rightarrow M_{3,2} \rightarrow A_{22} \rightarrow U_2 \rightarrow M_{2,3} \rightarrow A_{33} \rightarrow L_3 \rightarrow M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{11} \rightarrow U_1 \rightarrow$ the robot rotates $\rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow$ the robot rotates $\rightarrow L_1 \rightarrow M_{1,2} \rightarrow A_{23} \rightarrow L_2 \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rangle$ should be performed. With m_2 parallel PMs serving for Step 2, according to σ_{12} , the time for processing four wafers at Step 2 can be calculated using Equation (14):

$$\theta_2 = (\alpha_2 + d_2 + 2\beta_0 + 10\beta + 11\mu + v + \omega_{33} + \omega_{22} + \omega_{23} + \omega_{11} + \omega_{12})/m_2, n > 2 \quad (14)$$

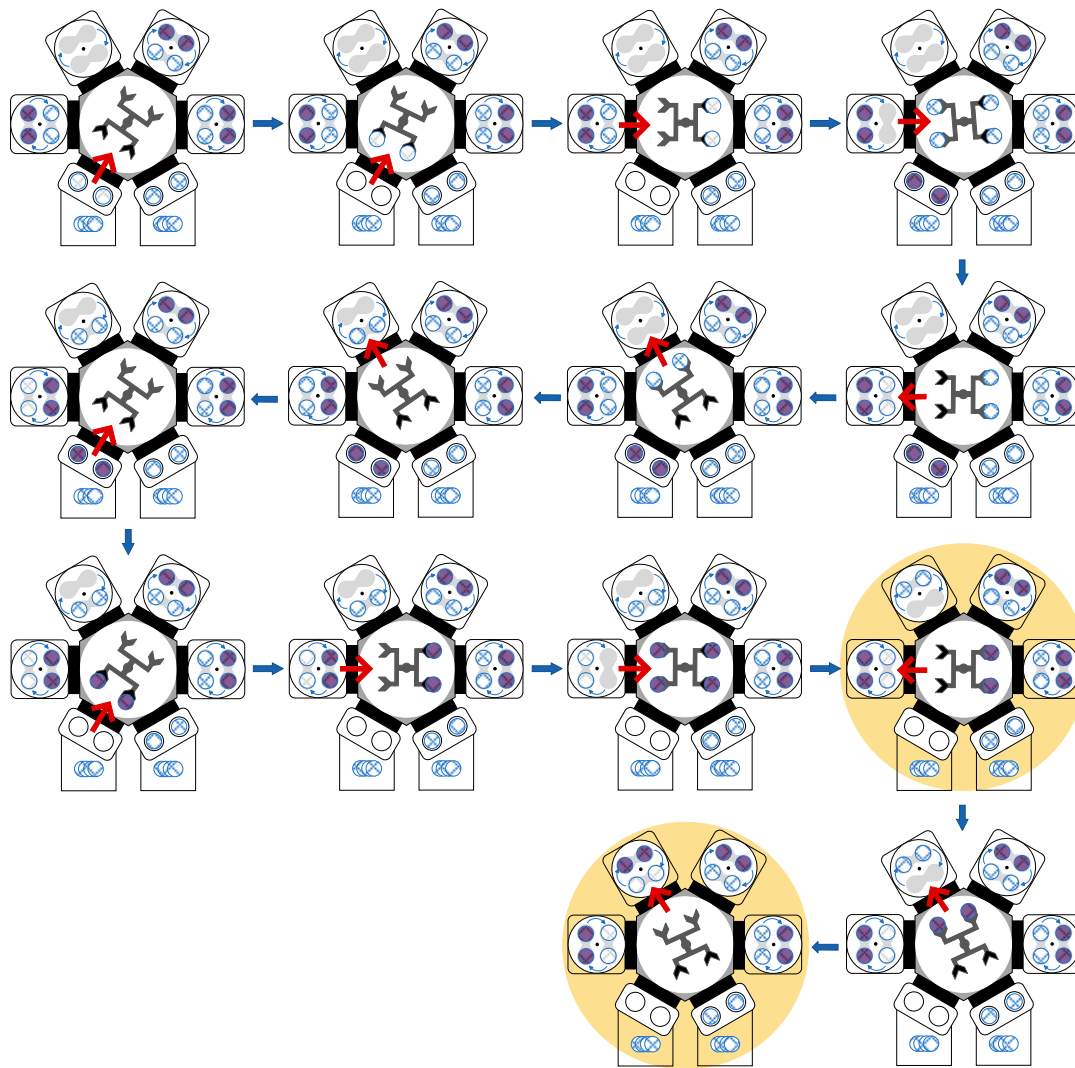


Figure 4. Description of robot movements under the OHTS strategy.

For Step 2, $n = 2$, to complete four wafers at a PM, the robot activity time is equivalent to that of Step 2 with $n > 2$ in the task sequence. However, the waiting time is not the same due to the absence of Step 3, which is A_{33} . With m_2 parallel PMs serving for Step 2, $n = 2$, the time for processing four wafers at Step 2 can be calculated using Equation (15).

$$\theta_2 = (\alpha_2 + d_2 + 2\beta_0 + 10\beta + 11\mu + v + \omega_{22} + \omega_{23} + \omega_{11} + \omega_{12})/m_2, n = 2 \quad (15)$$

For Step 1, to complete four wafers by a PM, task sequence $\sigma_{13} = \langle U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow \text{the internal chamber of the PM rotates} \rightarrow \text{wafer processing} \rightarrow \text{four wafers stay in the PM after processed (i.e., the delay time)} \rangle$ should be performed. With m_1 parallel PMs serving for Step 1, according to σ_{13} , the time for processing four wafers at Step 1 can be calculated using Equation (16):

$$\theta_1 = (\alpha_1 + d_1 + \beta_0 + 5\beta + 5\mu + v + \omega_{12})/m_1 \quad (16)$$

The robot-waiting activity A_{i2} , $i \in \mathbb{N}_n \setminus \{0\}$ is caused by the rotation of the internal chamber of a PM for Step i . Similar to OBS, the time taken for A_{i2} , $i \in \mathbb{N}_n \setminus \mathbb{N}_1$, is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{i,i+1} \rightarrow L_{i+1} \rightarrow M_{i+1,i} \rangle$. Thus, ω_{i2} , $i \in \mathbb{N}_n \setminus \mathbb{N}_1$ can be obtained using Equation (6).

Observing σ_{13} , after the robot performs a loading task at Step 1 for the first time so as to complete a robot rotation operation, the internal chamber of the PM starts to rotate. At the same time, the robot performs the task sequence $\langle M_{1,2} \rightarrow L_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rangle$. Thus, the time taken for A_{12} is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{1,2} \rightarrow L_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rangle$. Therefore, ω_{12} can be calculated using Equation (17):

$$\omega_{12} = \max(v - (\beta + \beta_0 + 3\mu), 0) \quad (17)$$

The robot-waiting activity A_{i3} , $i \in \mathbf{N}_n \setminus \{0\}$ is caused by the rotation of the internal chamber of a PM for Step i as well. Similar to OBS, the time taken for A_{i3} , $i \in \mathbf{N}_n \setminus \mathbf{N}_2$ is determined by the longest one between the time taken for a rotation operation of a PM for Step i and $\langle M_{i,i-1} \rightarrow A_{(i-1)2} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rangle$. Thus, ω_{i3} , $i \in \mathbf{N}_n \setminus \mathbf{N}_2$, can be obtained using Equation (7).

Observing σ_{12} , after the robot performs a loading task at Step 2 for the first time, the internal chamber of the PM starts to rotate. At the same time, the robot performs task sequence $\langle M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rangle$. Thus, the time taken for A_{23} is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rangle$. Therefore, Equation (18) is given to calculate ω_{23} :

$$\omega_{23} = \max(v - (2\beta + \beta_0 + 4\mu + \omega_{12}), 0) \quad (18)$$

Notice that A_{13} represents the robot-waiting activity at a PM for Step 1 before the robot loads wafers represented by $W_{1,34}$ into the PM. Observing σ_{13} , since the robot performs a swap operation at the PM such that these wafers are loaded into the PM, it implies that A_{13} is not necessary.

Moreover, similar to OBS, for Step i , $i \in \mathbf{N}_n \setminus \mathbf{N}_2$, the wafer sojourn time of $W_{i,12}$ and $W_{i,34}$ can be obtained by (9) and (10), respectively. For Step 2, observing σ_{12} , the time taken for task sequence $\langle M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rightarrow A_{23} \rightarrow L_2 \rightarrow \text{the internal chamber of the PM rotates} \rightarrow \text{wafer processing} \rightarrow \text{four wafers stay in the PM after processed (i.e., the delay time)} \rangle$ equals the wafer sojourn time of $W_{2,12}$. Also, the time taken for task sequence $\langle \text{the internal chamber of the PM rotates} \rightarrow \text{wafer processing} \rightarrow \text{four wafers stay in the PM after processed (i.e., the delay time)} \rightarrow U_2 \rightarrow M_{2,3} \rightarrow L_3 \rightarrow M_{3,2} \rightarrow A_{22} \rangle$ equals the wafer sojourn time of $W_{2,34}$. Then, Equation (19) is given to obtain $\tau_{2,12}$, while $\tau_{2,34}$ can be obtained by (10).

$$\tau_{2,12} = \alpha_2 + d_2 + v + \beta_0 + 3\beta + 4\mu + \omega_{12} + \omega_{23} \quad (19)$$

For Step 1, observing σ_{13} , the time taken for task sequence $\langle M_{1,2} \rightarrow L_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow \text{the internal chamber of the PM rotates} \rightarrow \text{wafer processing} \rightarrow \text{four wafers stay in the PM after processed (i.e., the delay time)} \rangle$ equals the wafer sojourn time of $W_{1,12}$, and the time taken for task sequence $\langle \text{the internal chamber of the PM rotates} \rightarrow \text{wafer processing} \rightarrow \text{four wafers stay in the PM after processed (i.e., the delay time)} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rangle$ equals the wafer sojourn time of $W_{1,34}$. Therefore, we have the following Equation:

$$\tau_{1,12} = \tau_{1,34} = \alpha_1 + d_1 + v + 3\beta + \beta_0 + 4\mu + \omega_{12} \quad (20)$$

For OHTS, with the robot-waiting activities being considered, σ_3 and σ_4 can be rewritten as $\langle U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow A_{(n-1)1} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-1} \rightarrow A_{(n-1)2} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow A_{n3} \rightarrow L_n \rightarrow M_{n,n-2} \rightarrow \dots \rightarrow M_{4,2} \rightarrow A_{21} \rightarrow U_2 \rightarrow M_{2,3} \rightarrow L_3 \rightarrow M_{3,2} \rightarrow A_{22} \rightarrow U_2 \rightarrow M_{2,3} \rightarrow A_{33} \rightarrow L_3 \rangle$ and $\langle M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{11} \rightarrow \text{performing a swap operation at Step 1} \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow \text{performing a swap operation at Step 1} \rightarrow M_{1,2} \rightarrow A_{23} \rightarrow$

$L_2 \rightarrow M_{2,n}$, respectively. Then, under OHTS, the robot cycle time ψ_2 formed by σ_3 and σ_4 together can be calculated as follows:

$$\psi_2 = 4(n+1)\mu + 2(2n+1)\beta + 2\beta_0 + \sum_1^n \omega_{i1} + \sum_1^n \omega_{i2} + \sum_1^n \omega_{i3} \quad (21)$$

3.3. Temporal Properties Under TBS

Under the steady state, to complete four wafers at a PM for Step i , $i \in \mathbf{N}_n \setminus \mathbf{N}_1$, by TBS, with the robot-waiting activities being considered, task sequence $\sigma_{14} = \langle U_i \rightarrow M_{i,i+1} \rightarrow L_{i+1} \rightarrow M_{i+1,i-1} \rightarrow A_{(i-1)1} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rightarrow L_i \rightarrow M_{i,i-2} \rightarrow \dots \rightarrow M_{3,1} \rightarrow A_{11} \rightarrow U_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{i+1,i-1} \rightarrow A_{(i-1)2} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rightarrow L_i \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rangle$ should be performed. Figure 5 shows the robot task sequence under the TBS strategy with part of the backward process omitted, in the figure the meaning of the colored arrows is same as that in Figure 3. The steps with a yellow circular background similarly indicate the internal chamber rotation of the PM before the processing step. With m_i parallel PMs serving for Step i , according to σ_{14} , the time for processing four wafers can be calculated with Equation (22):

$$\theta_i = \left[\alpha_i + d_i + (2n+5)\beta + \beta_0 + (2n+5)\mu + v + \sum_1^{i-1} \omega_{i1} + \sum_{i-1}^n \omega_{i2} \right] / m_i, i \in \mathbf{N}_n \setminus \mathbf{N}_1 \quad (22)$$

Similarly, to complete four wafers by a PM for Step 1, by TBS, with the robot-waiting activities being considered, task sequence $\sigma_{15} = \langle U_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rangle$ should be performed. With m_1 parallel PMs serving for Step 1, according to σ_{15} , the time for processing four wafers can be calculated using Equation (23):

$$\theta_1 = \left[\alpha_1 + d_1 + (2n+4)\beta + 2\beta_0 + (2n+5)\mu + v + \sum_1^n \omega_{i2} \right] / m_1 \quad (23)$$

In σ_{14} , A_{i2} is also caused by the rotation of the internal chamber of a PM for Step i . After the robot performs a loading task at the PM in σ_{14} for the first time, the internal chamber of the PM starts to rotate. At the same time, the robot performs task sequence $\langle M_{i,i-2} \rightarrow A_{(i-2)1} \rightarrow U_{i-2} \rightarrow M_{i-2,i-1} \rightarrow L_{i-1} \rightarrow M_{i-1,i-3} \rightarrow \dots \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{i+2,i} \rangle$. It follows from this sequence that A_{12} and A_{22} are different, implying that A_{i2} is dependent on n . Thus, we divide it into two cases: $n = 2$ and $n > 2$. If $n > 2$, A_{i2} , $i \in \{3, 4, \dots, n-1\}$ should be performed. And the time taken for A_{i2} is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{i,i-2} \rightarrow A_{(i-2)1} \rightarrow U_{i-2} \rightarrow M_{i-2,i-1} \rightarrow L_{i-1} \rightarrow M_{i-1,i-3} \rightarrow \dots \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{i+2,i} \rangle$. When $i = 2$ and $n > 2$, the time taken for A_{22} is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{i+2,i} \rightarrow A_{i2} \rightarrow U_i \rightarrow M_{i,i+1} \rightarrow L_{i+1} \rightarrow M_{i+1,i-1} \rightarrow \dots \rightarrow M_{5,3} \rightarrow A_{32} \rightarrow U_3 \rightarrow M_{3,4} \rightarrow L_4 \rightarrow M_{4,2} \rangle$. When $n = 2$, A_{22} should be performed. And the time taken for A_{22} is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,2} \rangle$. Thus, we have the following Equations:

$$\omega_{i2} = \max(v - [(2n-1)\mu + (2n-3)\beta + \beta_0 + \sum_1^{i-2} \omega_{i1} + \sum_{i+1}^n \omega_{i2}], 0), i \in \{3, 4, \dots, n-1\} \text{ and } n > 3 \quad (24)$$

$$\omega_{22} = \max(v - [(2n-1)\mu + (2n-3)\beta + \beta_0 + \sum_3^n \omega_{i2}], 0), n > 2 \quad (25)$$

$$\omega_{22} = \max(v - (3\mu + \beta + \beta_0), 0), n = 2 \quad (26)$$

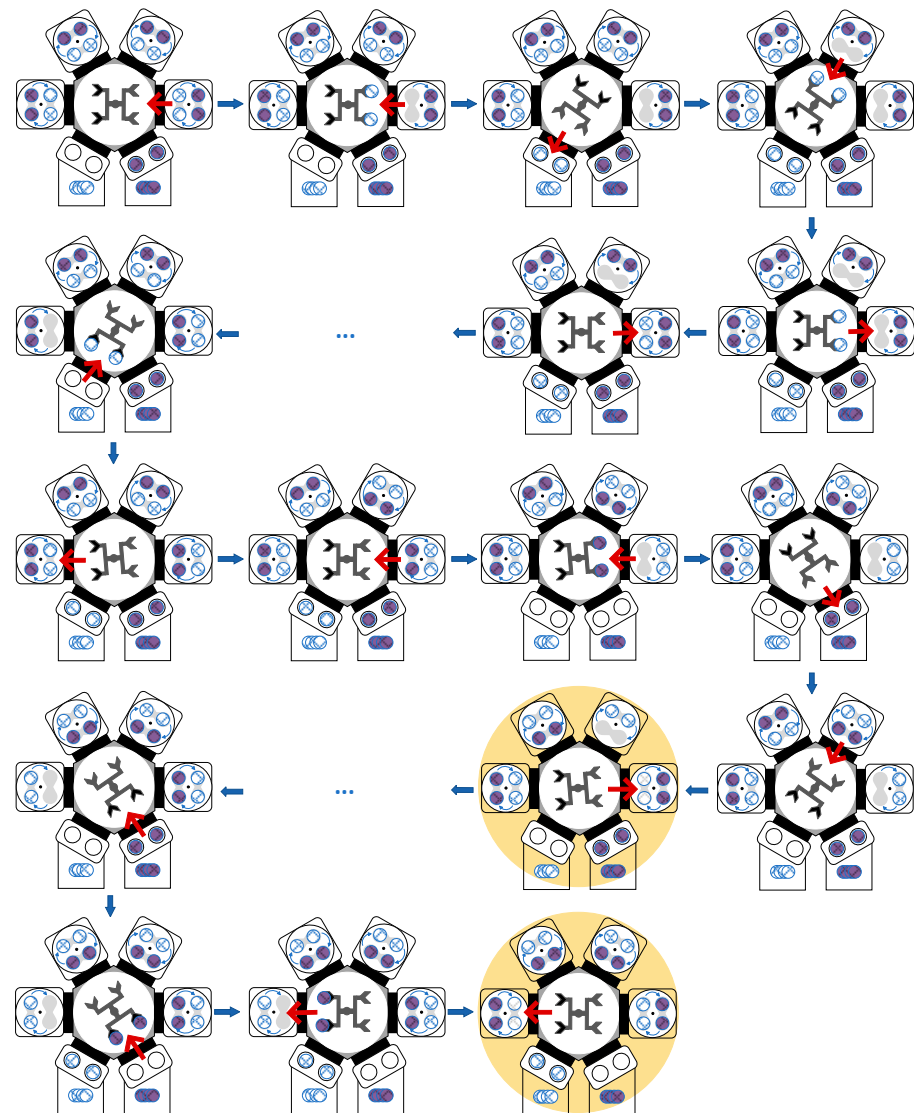


Figure 5. Description of robot movements under the TBS strategy.

When $i = n$ and $n > 2$, the robot performs task sequence $\langle M_{n,n-2} \rightarrow A_{(n-2)1} \rightarrow U_{n-2} \rightarrow M_{n-2,n-1} \rightarrow L_{n-1} \rightarrow M_{n-1,n-3} \rightarrow \dots \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rangle$. Then, A_{n2} should be performed. And the time taken for A_{n2} is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{n,n-2} \rightarrow A_{(n-2)1} \rightarrow U_{n-2} \rightarrow M_{n-2,n-1} \rightarrow L_{n-1} \rightarrow M_{n-1,n-3} \rightarrow \dots \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rangle$. Then, we can obtain ω_{n2} as follows:

$$\omega_{n2} = \max(v - [(2n - 1)\mu + (2n - 3)\beta + \beta_0 + \sum_1^{n-2} \omega_{i1}], 0), n > 2 \quad (27)$$

Similarly, we have

$$\omega_{12} = \max(v - [(2n - 1)\mu + (2n - 2)\beta + \sum_2^n \omega_{i2}], 0). \quad (28)$$

According to σ_{14} , the time taken for task sequence $\langle M_{i,i-2} \rightarrow A_{(i-2)1} \rightarrow U_{i-2} \rightarrow M_{i-2,i-1} \rightarrow L_{i-1} \rightarrow M_{i-1,i-3} \rightarrow \dots \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{i+1,i-1} \rightarrow A_{(i-1)2} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rightarrow L_i \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM

after processed (i.e., the delay time)) equals the wafer sojourn time of $W_{i,12}$, $i \in \mathbf{N}_n \setminus \mathbf{N}_2$. Furthermore, the time taken for the task sequence (the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rightarrow U_i \rightarrow M_{i,i+1} \rightarrow L_{i+1} \rightarrow M_{i+1,i-1} \rightarrow A_{(i-1)1} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rightarrow L_i \rightarrow M_{i,i-2} \rightarrow \dots \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{i+2,i} \rightarrow A_{i2}$) equals the wafer sojourn time of $W_{i,34}$, $i \in \mathbf{N}_n \setminus \mathbf{N}_1$. Hence, we have the following:

$$\tau_{i,12} = \alpha_i + d_i + v + (2n+1)\beta + \beta_0 + (2n+2)\mu + \sum_{i=1}^{i-2} \omega_{i1} + \sum_{i=1}^n \omega_{i2}, \quad i \in \mathbf{N}_n \setminus \mathbf{N}_2, n > 2 \quad (29)$$

$$\tau_{i,34} = \alpha_i + d_i + v + (2n+1)\beta + \beta_0 + (2n+2)\mu + \sum_{i=1}^{i-1} \omega_{i1} + \sum_{i=1}^n \omega_{i2}, \quad i \in \mathbf{N}_n \setminus \mathbf{N}_1 \quad (30)$$

For Step 2, the time taken for task sequence ($M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rightarrow \dots \rightarrow M_{i+1,i-1} \rightarrow A_{(i-1)2} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rightarrow L_i \rightarrow M_{i,i-1} \rightarrow \dots \rightarrow M_{3,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time)) equals the wafer sojourn time of $W_{2,12}$ that can be calculated with Equation (31), and the wafer sojourn time of $W_{2,34}$ is the same as that of $W_{2,12}$. We have the following:

$$\tau_{2,12} = \tau_{2,34} = \alpha_2 + d_2 + v + (2n+1)\beta + \beta_0 + (2n+2)\mu + \sum_{i=1}^n \omega_{i2} \quad (31)$$

According to σ_{15} , the time taken for task sequence ($M_{1,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow A_{(n-1)2} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-2} \rightarrow \dots \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time)) equals the wafer sojourn time of $W_{1,12}$. Furthermore, the time taken for task sequence (the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rightarrow U_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{3,1} \rightarrow A_{12}$) equals the wafer sojourn time of $W_{1,34}$. Based on the above analysis, we show $\tau_{1,12} = \tau_{1,34}$, and we have the following:

$$\tau_{1,12} = \tau_{1,34} = \alpha_1 + d_1 + v + (2n+1)\beta + \beta_0 + (2n+2)\mu + \sum_{i=1}^n \omega_{i2} \quad (32)$$

For TBS, with the robot-waiting activities being considered, σ_5 can be rewritten as $\sigma_5^1 = \langle U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow A_{(n-1)1} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-2} \rightarrow \dots \rightarrow M_{3,1} \rightarrow A_{11} \rightarrow U_1 \rightarrow M_{1,2} \rightarrow L_2 \rangle$ and $\sigma_5^2 = \langle U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow A_{(n-1)2} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-2} \rightarrow \dots \rightarrow M_{3,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow M_{1,2} \rightarrow L_2 \rangle$; while σ_6 can be rewritten as $\sigma_6^1 = \langle M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rightarrow A_{n2} \rangle$ and $\sigma_6^2 = \langle M_{2,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow L_1 \rightarrow M_{1,n} \rightarrow A_{n1} \rangle$. Then, under TBS, the robot cycle time ψ_3 taken by σ_5^1 , σ_5^2 , σ_6^1 , and σ_6^2 together can be obtained as follows:

$$\psi_3 = 4(n+1)\mu + 2(2n+1)\beta + 2\beta_0 + \sum_{i=1}^n \omega_{i1} + \sum_{i=1}^n \omega_{i2} \quad (33)$$

3.4. Temporal Properties Under THTS

Similar to TBS, to complete four wafers by a PM for Step i , $i \in \mathbf{N}_n \setminus \mathbf{N}_2$ and $n > 2$, by THTS, task sequence $\sigma_{16} = \langle U_i \rightarrow M_{i,i+1} \rightarrow L_{i+1} \rightarrow M_{i+1,i-1} \rightarrow A_{(i-1)1} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rightarrow L_i \rightarrow M_{i,i-2} \rightarrow \dots \rightarrow M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{11} \rightarrow U_1 \rightarrow$ the robot rotates $\rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{i+1,i-1} \rightarrow A_{(i-1)2} \rightarrow U_{i-1} \rightarrow M_{i-1,i} \rightarrow L_i \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time)) should be performed. Figure 6 presents the robot task sequence under the THTS strategy with part of the backward process omitted, in the figure, the meaning of the colored arrows is same as that in Figure 3. The steps with a yellow circular background similarly indicate the internal chamber rotation of the PM before the processing step. Thus, with m_i parallel PMs serving for Step i , according to σ_{16} , the time for processing four wafers at Step i , $i \in \mathbf{N}_n \setminus \mathbf{N}_2$ and $n > 2$, can be calculated with Equation (34):

$$\theta_i = [\alpha_i + d_i + (2n + 5)\beta + \beta_0 + (2n + 5)\mu + v + \sum_1^{i-1} \omega_{i1} + \sum_{i-1}^n \omega_{i2}] / m_i, i \in \mathbf{N}_n \setminus \mathbf{N}_2 \text{ and } n > 2 \quad (34)$$

For Step 2, $n > 2$, to complete four wafers by a PM, task sequence $\sigma_{17} = \langle U_2 \rightarrow M_{2,3} \rightarrow L_3 \rightarrow M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{11} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow \text{the internal chamber of the PM rotates} \rightarrow \text{wafer processing} \rightarrow \text{four wafers stay in the PM after processed (i.e., the delay time)} \rangle$ should be performed. When $n = 2$, for Step 2, to complete four wafers by a PM, the task sequence $\sigma_{18} = \langle U_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow \text{the robot rotates} \rightarrow L_0 \rightarrow M_{0,1} \rightarrow A_{11} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow \text{the internal chamber of the PM rotates} \rightarrow U_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow \text{the robot rotates} \rightarrow L_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow \text{the internal chamber of the PM rotates} \rightarrow \text{wafer processing} \rightarrow \text{four wafers stay in the PM after processed (i.e., the delay time)} \rangle$ should be performed. With m_2 parallel PMs serving for Step 2, according to σ_{17} and σ_{18} , the time for processing four wafers at Step 2 can be calculated using Equations (35) and (36):

$$\theta_2 = [\alpha_2 + d_2 + (2n + 6)\beta + 2\beta_0 + (2n + 7)\mu + v + \omega_{11} + \sum_1^n \omega_{i2}] / m_2, n > 2 \quad (35)$$

$$\theta_2 = (\alpha_2 + d_2 + 10\beta + 2\beta_0 + 10\mu + 2v + \omega_{11} + \omega_{12}) / m_2, n = 2 \quad (36)$$

For Step 1, $n > 2$, to complete four wafers by a PM, task sequence $\sigma_{19} = \langle U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow \text{the internal chamber of the PM rotates} \rightarrow \text{wafer processing} \rightarrow \text{four wafers stay in the PM after processed (i.e., the delay time)} \rangle$ should be performed. When $n = 2$, for Step 1, to complete four wafers by a PM, the task sequence $\sigma_{20} = \langle U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow \text{the internal chamber of the PM rotates} \rightarrow U_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow \text{the robot rotates} \rightarrow L_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow \text{the internal chamber of the PM rotates} \rightarrow \text{wafer processing} \rightarrow \text{four wafers stay in the PM after processed (i.e., the delay time)} \rangle$ should be performed. With m_1 parallel PMs serving for Step 1, according to σ_{19} and σ_{20} , the time for processing four wafers at Step 1 can be calculated with Equations (37) and (38):

$$\theta_1 = [\alpha_1 + d_1 + (2n + 3)\beta + \beta_0 + (2n + 3)\mu + v + \sum_1^n \omega_{i2}] / m_1, n > 2 \quad (37)$$

$$\theta_1 = (\alpha_1 + d_1 + 7\beta + \beta_0 + 6\mu + 2v + \omega_{12}) / m_1, n = 2 \quad (38)$$

By observing σ_{16} , the time taken for the robot-waiting activity $A_{i2}, i \in \{3, \dots, n-1\}$ and $n > 3$, is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{i,i-2} \rightarrow A_{(i-2)1} \rightarrow U_{i-2} \rightarrow M_{i-2,i-1} \rightarrow L_{i-1} \rightarrow M_{i-1,i-3} \rightarrow \dots \rightarrow M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{11} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{i+2,i} \rangle$. When $n > 2$, A_{i2} is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{n,n-2} \rightarrow A_{(n-2)1} \rightarrow U_{n-2} \rightarrow M_{n-2,n-1} \rightarrow L_{n-1} \rightarrow M_{n-1,n-3} \rightarrow \dots \rightarrow M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{11} \rightarrow U_1 \rightarrow \text{the robot rotates} \rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,n} \rangle$. Thus, $\omega_{i2}, i \in \{3, \dots, n-1\}$ and $n > 3$, and $\omega_{n2}, n > 2$ can be obtained with Equations (39) and (40):

$$\omega_{i2} = \max(v - [(2n-1)\mu + (2n-3)\beta + \beta_0 + \sum_1^{i-2} \omega_{i1} + \sum_{i+1}^n \omega_{i2}], 0), i \in \{3, 4, \dots, n-1\} \text{ and } n > 3 \quad (39)$$

$$\omega_{n2} = \max(v - [(2n-1)\mu + (2n-3)\beta + \beta_0 + \sum_1^{n-2} \omega_{i1}], 0), n > 2 \quad (40)$$

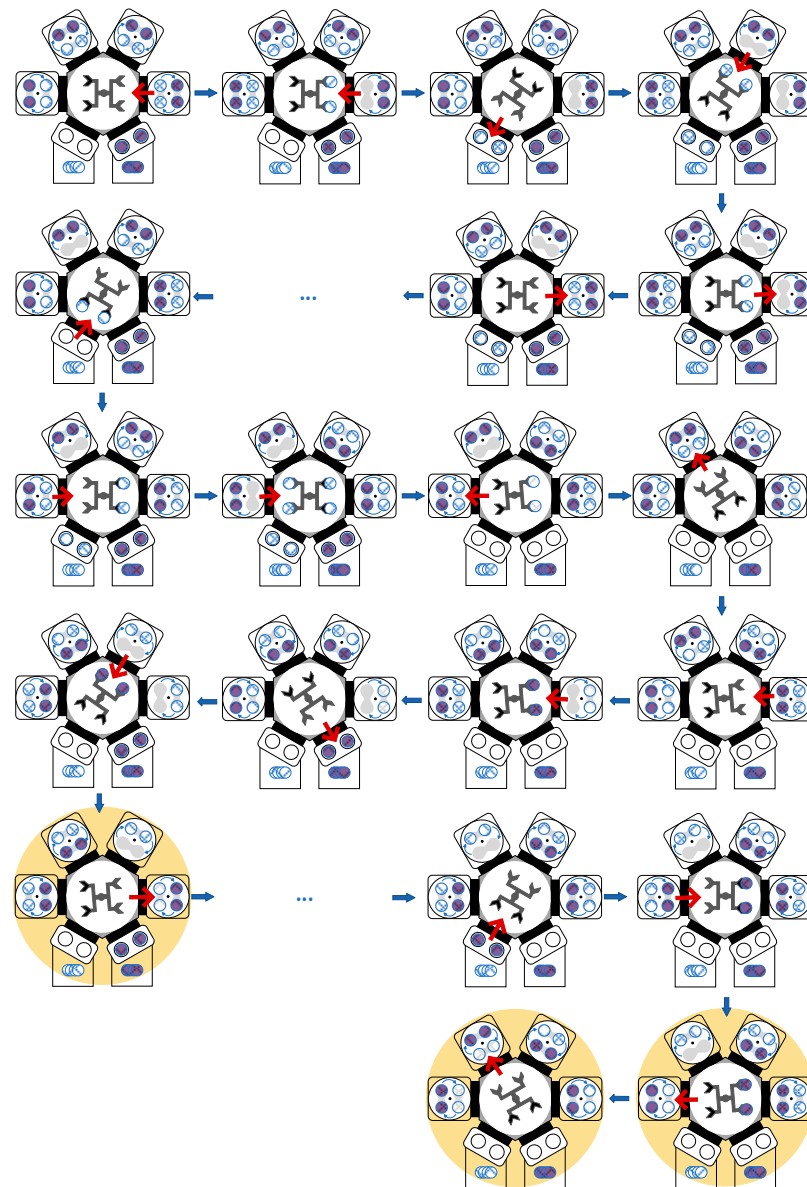


Figure 6. Description of robot movements under the THTS strategy.

Observing σ_{17} and σ_{18} , after the robot performs a loading task at Step 2 for the first time so as to complete a robot rotation operation, the internal chamber of the PM starts to rotate. Hence, the time taken for the robot-waiting activity A_{22} , $n > 2$, is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{2,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{5,3} \rightarrow A_{32} \rightarrow U_3 \rightarrow M_{3,4} \rightarrow L_4 \rightarrow M_{4,2} \rangle$. When $n = 2$, the robot just needs to wait for the completion of the internal rotation of the PM. Therefore, ω_{22} should be calculated using Equations (41) and (42):

$$\omega_{22} = \max(v - [(2n - 3)\mu + 2(n - 2)\beta + \sum_3^n \omega_{i2}], 0), n > 2 \quad (41)$$

$$\omega_{22} = v, n = 2 \quad (42)$$

Observing σ_{19} and σ_{20} , after the robot performs a loading task at Step 1 for the first time so as to complete a robot rotation operation, the internal chamber of the PM starts to rotate. Hence, the time taken for robot-waiting activity A_{12} , $n > 2$, is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{1,2} \rightarrow L_2 \rightarrow M_{2,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{5,3} \rightarrow A_{32} \rightarrow U_3 \rightarrow M_{3,4} \rightarrow L_4 \rightarrow M_{4,2} \rightarrow A_{22} \rightarrow U_2 \rightarrow M_{2,3} \rightarrow L_3 \rightarrow M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rangle$. When $n = 2$, the time taken

for robot waiting activity A_{12} is determined by the longest one between the time taken for a rotation operation of the PM and $\langle M_{1,2} \rightarrow L_2 \rightarrow A_{22} \rightarrow U_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow$ the robot rotates $\rightarrow L_0 \rightarrow M_{0,1} \rangle$. Thus, ω_{12} should be calculated with Equations (43) and (44):

$$\omega_{12} = \max(v - [(2n+1)\mu + (2n-1)\beta + \sum_2^n \omega_{i2}], 0), n > 2 \quad (43)$$

$$\omega_{12} = \max(v - (4\mu + 3\beta + \beta_0 + v), 0), n = 2 \quad (44)$$

Similar to TBS, for Step i , $i \in \mathbf{N}_n \setminus \mathbf{N}_2$, the wafer sojourn time of $W_{i,12}$ and $W_{i,34}$ can be obtained with (29) and (30), respectively. For Step 2, $n > 2$, the wafer sojourn time of $W_{2,12}$ and $W_{2,34}$ can be obtained using (31) and (32), respectively. For Step 2, $n = 2$, the time taken for task sequence $\langle$ the internal chamber of the PM rotates $\rightarrow U_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow$ the robot rotates $\rightarrow L_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow$ the robot rotates $\rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rangle$ equals the wafer sojourn time of $W_{2,12}$. Furthermore, the time taken for task sequence $\langle$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rightarrow U_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow$ the robot rotates $\rightarrow L_0 \rightarrow M_{0,1} \rightarrow A_{11} \rightarrow U_1 \rightarrow$ the robot rotates $\rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow$ the internal chamber of the PM rotates $\rangle$ equals the wafer sojourn time of $W_{2,34}$. Therefore, the wafer sojourn time of $W_{2,12}$ and $W_{2,34}$ can be calculated using Equations (45) and (46):

$$\tau_{2,12} = \alpha_2 + d_2 + 2v + 5\beta + \beta_0 + 5\mu + \omega_{12}, n = 2 \quad (45)$$

$$\tau_{2,34} = \alpha_2 + d_2 + 2v + 5\beta + \beta_0 + 5\mu + \omega_{11}, n = 2 \quad (46)$$

For Step 1, $n > 2$, the time taken for the task sequence $\langle M_{1,2} \rightarrow L_2 \rightarrow M_{2,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow$ the robot rotates $\rightarrow L_1 \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rangle$ equals the wafer sojourn time of $W_{1,12}$. Furthermore, the time taken for the task sequence $\langle$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rightarrow U_1 \rightarrow$ the robot rotates $\rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,n} \rightarrow A_{n2} \rightarrow U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow \dots \rightarrow M_{4,2} \rightarrow A_{4,22} \rightarrow U_2 \rightarrow M_{2,3} \rightarrow L_3 \rightarrow M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rangle$ equals the wafer sojourn time of $W_{1,34}$. Based on the above analysis, we obtain that $\tau_{1,12} = \tau_{1,34}$, $n > 2$, and they can be calculated as follows:

$$\tau_{1,12} = \tau_{1,34} = \alpha_1 + d_1 + v + (2n+1)\beta + \beta_0 + (2n+2)\mu + \sum_1^n \omega_{i2}, n > 2 \quad (47)$$

For Step 1, $n = 2$, the time taken for task sequence $\langle M_{1,2} \rightarrow L_2 \rightarrow$ the internal chamber of the PM rotates $\rightarrow U_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow$ the robot rotates $\rightarrow L_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow U_1 \rightarrow$ the robot rotates $\rightarrow L_1 \rightarrow$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rangle$ equals the wafer sojourn time of $W_{1,12}$. Furthermore, the time taken for task sequence $\langle$ the internal chamber of the PM rotates $\rightarrow$ wafer processing $\rightarrow$ four wafers stay in the PM after processed (i.e., the delay time) $\rightarrow U_1 \rightarrow$ the robot rotates $\rightarrow L_1 \rightarrow M_{1,2} \rightarrow L_2 \rightarrow$ the internal chamber of the PM rotates $\rightarrow U_2 \rightarrow M_{2,0} \rightarrow U_0 \rightarrow$ the robot rotates $\rightarrow L_0 \rightarrow M_{0,1} \rightarrow A_{12} \rangle$ equals the sojourn time of $W_{1,34}$. Therefore, the wafer sojourn time of $W_{1,12}$ and $W_{1,34}$ can be calculated using Equation (48):

$$\tau_{1,12} = \tau_{1,34} = \alpha_1 + d_1 + 2v + 5\beta + \beta_0 + 5\mu + \omega_{12}, n = 2 \quad (48)$$

For THTS, with the robot-waiting activities being considered, σ_7 can be rewritten as $\sigma_7^1 = \langle U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow A_{(n-1)1} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-2} \rightarrow \dots \rightarrow M_{4,2} \rightarrow A_{21} \rightarrow U_2 \rightarrow M_{2,3} \rightarrow L_3 \rangle$ and $\sigma_7^2 = \langle U_n \rightarrow M_{n,n+1} \rightarrow L_{n+1} \rightarrow M_{n+1,n-1} \rightarrow A_{(n-1)2} \rightarrow U_{n-1} \rightarrow M_{n-1,n} \rightarrow L_n \rightarrow M_{n,n-2} \rightarrow \dots \rightarrow M_{4,2} \rightarrow A_{22} \rightarrow U_2 \rightarrow M_{2,3} \rightarrow L_3 \rangle$, while σ_8 can be rewritten as $\sigma_8^1 = \langle M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{11} \rightarrow$ performing a swap operation at Step 1 $\rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,n} \rightarrow A_{n1} \rangle$ and $\sigma_8^2 = \langle M_{3,0} \rightarrow U_0 \rightarrow M_{0,1} \rightarrow A_{12} \rightarrow$ performing a swap

operation at Step 1 $\rightarrow M_{1,2} \rightarrow L_2 \rightarrow M_{2,n} \rightarrow A_{n2}$). Then, under THTS, the robot cycle time ψ_4 taken by σ_7^1 , σ_7^2 , σ_8^1 , and σ_8^2 together can be obtained as follows:

$$\psi_4 = 4(n+1)\mu + 2(2n+1)\mu + 2\mu_0 + \sum_1^n \omega_{i1} + \sum_1^n \omega_{i2} \quad (49)$$

3.5. Linear Programs for Solution

By any one of OBS, OHTS, TBS, and THTS, if an SF³–CT operates under the steady state, in a robot task cycle, each step completes four wafers, and four completed wafers are returned to LLs. Also, the time taken to complete four wafers at each step equals the robot cycle time. Thus, we have Propositions 1–4:

Proposition 1. For an SF³–CT, if the tool is operated by OBS under the steady state, the robot cycle time equals the time for completing four wafers at each step, i.e., Equations (50)–(52) hold:

$$\psi_1 = \theta_1 \quad (50)$$

$$\psi_1 = \theta_i, i \in \mathbf{N}_n \setminus \mathbf{N}_1 \text{ and } n > 2 \quad (51)$$

$$\psi_1 = \theta_n \quad (52)$$

where $\theta_n, \theta_i, i \in \mathbf{N}_n \setminus \mathbf{N}_1$ and $n > 2$, and θ_1 are obtained by (3)–(5), respectively.

Proposition 2. For an SF³–CT, if the tool is operated by OHTS under the steady state, the robot cycle time equals the time for completing four wafers at each step, i.e., Equations (53)–(56) hold:

$$\psi_2 = \theta_1 \quad (53)$$

$$\psi_2 = \theta_2, n \geq 2 \quad (54)$$

$$\psi_2 = \theta_i, i \in \{2, 3, \dots, n-1\} \text{ and } n > 3 \quad (55)$$

$$\psi_2 = \theta_n \quad (56)$$

where $\theta_2, n > 2$; $\theta_2, n = 2$; θ_1 ; $\theta_i, i \in \{2, 3, \dots, n-1\}$ and $n > 3$; and θ_n are obtained by (14)–(16), (4), and (3), respectively.

Proposition 3. For an SF³–CT, if the tool is operated by TBS under the steady state, the robot cycle time equals the time for completing four wafers at each step, i.e., Equation (57) holds:

$$\psi_3 = \theta_1 = \theta_i \quad (57)$$

where θ_1 and θ_i are obtained by (23) and (22), respectively.

Proposition 4. For an SF³–CT, if the tool is operated by THTS under the steady state, the robot cycle time equals the time for completing four wafers at each step, i.e., Equations (58)–(60) hold:

$$\psi_4 = \theta_1 = \theta_2, n = 2 \quad (58)$$

$$\psi_4 = \theta_1 = \theta_2, n > 2 \quad (59)$$

$$\psi_4 = \theta_i, i \in \mathbf{N}_n \setminus \mathbf{N}_2 \text{ and } n > 2 \quad (60)$$

where $\theta_i, i \in \mathbf{N}_n \setminus \mathbf{N}_2$ and $n > 2$; $\theta_2, n > 2$; $\theta_2, n = 2$; $\theta_1, n = 2$; and $\theta_1, n > 2$, are obtained by (34)–(38), respectively.

Also to meet RTCs, (1) and (2) should be held when any strategy is applied. Note that A_{i2} and A_{i3} occur during the processes for unloading wafers from a PM and loading wafers into a PM, respectively. Thus, in terms of the satisfaction of RTCs, it is desired that ω_{i2}

and ω_{i3} (the time taken by A_{i2} and A_{i3}) should be as short as possible. Hence, with this observation in mind, according to the above analysis, ω_{i2} and ω_{i3} are already determined.

Due to that, for a steady-state schedule, the cycle time for each processing step is the same, and it should be equal to the robot cycle time; when the robot-waiting time at all the steps is determined, a schedule is obtained. Note that the robot-waiting time is a continuous variable. The problem becomes a continuous optimization problem but not a discrete optimization problem. Hence, given a strategy, to find a schedule is to determine ω_{i1} (the time taken by A_{i1}). Based on Propositions 1–4, we establish four linear programs for different strategies to find optimal and feasible schedules with the objective of minimizing the robot cycle time, i.e., the system cycle time. Based on the property of a schedule, when a schedule is found, the robot's time occupancy is known.

Linear Program 1 (LP1). For an SF^3 -CT, if the tool is operated by OBS under the steady state, a linear program is established to obtain a schedule as follows:

Minimize ψ_1

Subject to: (1)–(13), (50)–(52).

In LP1, Constraints (1) and (2) guarantee that the wafer sojourn time is within the permissible time windows. Constraints (3)–(5) calculate the time required to complete four wafers at each step from Steps 1 to n . Then, Constraints (6)–(8) calculate the time for robot-waiting activities A_{i2} and A_{i3} at each step. Constraints (9)–(12) determine the sojourn time of $W_{i,12}$ and $W_{i,34}$. Equation (15) obtains the robot cycle time under OBS. Constraints (50)–(52) come from Proposition 1, and we can obtain ω_{i1} and ψ_1 .

Linear Program 2 (LP2). For an SF^3 -CT, if the tool is operated by OHTS under the steady state, a linear program is established to obtain a schedule as follows:

Minimize ψ_2

Subject to: (1)–(3),

(14)–(21), (53)–(56)

(4) where $i \in \{2, 3, \dots, n-1\}$ and $n > 3$,

(6) and (10) where $i \in \mathbf{N}_n \setminus \mathbf{N}_1$, and

(7) and (9) where $i \in \mathbf{N}_n \setminus \mathbf{N}_2$.

In LP2, similar to LP1, Constraints (1) and (2) guarantee that the RTCs are satisfied. Constraints (3), (4), (14), and (15) calculate the time required to complete four wafers at each step from Steps 1 to n . Then, Constraints (6), (7), (17), and (18) calculate the time for robot-waiting activities A_{i2} and A_{i3} at each step. Constraints (9), (10), (19), and (20) determine the sojourn time of $W_{i,12}$ and $W_{i,34}$. Equation (20) obtains the robot cycle time under OHTS, i.e., the system cycle time. Constraints (53)–(56) come from Proposition 2, and we can obtain ω_{i1} and ψ_2 .

Linear Program 3 (LP3). For an SF^3 -CT, if the tool is operated by TBS under the steady state, a linear program is established to obtain a schedule as follows.

Minimize ψ_3

Subject to: (1), (2), (22)–(33), (57)

Similar to LP1 and LP2, Constraints (1) and (2) guarantee that the RTCs are satisfied. Constraints (22) and (23) calculate the time required to complete four wafers at each step from Steps 1 to n . Then, Constraints (24)–(28) calculate the time for robot waiting

activities A_{i2} at each step. Constraints (29)–(32) determine the sojourn time of $W_{i,12}$ and $W_{i,34}$. Equation (33) obtains the robot cycle time under TBS. Constraint (57) comes from Proposition 3, and we can obtain ω_{i1} and ψ_3 .

Linear Program 4 (LP4). For an SF^3 –CT, if the tool is operated by THTS under the steady state, a linear program is established to obtain a schedule as follows.

$$\begin{aligned} & \text{Minimize } \psi_4 \\ & \text{Subject to: (1), (2),} \\ & \quad (34)–(49), (58)–(60), \\ & \quad (29) \text{ where } i \in \mathbf{N}_n \setminus \mathbf{N}_2 \text{ and } n > 2, \\ & \quad (30) \text{ where } i \in \mathbf{N}_n \setminus \mathbf{N}_1 \text{ and } n > 2, \text{ and} \\ & \quad (31) \text{ in which } n > 2. \end{aligned}$$

Similar to the linear programs above, Constraints (1) and (2) guarantee that the RTCs are satisfied. Constraints (34)–(38) calculate the time required to complete four wafers at each step from Steps 1 to n . Then, Constraints (39)–(44) calculate the time for robot-waiting activities A_{i2} at each step. Constraints (29)–(31) and (45)–(48) determine the sojourn time of $W_{i,12}$ and $W_{i,34}$. Equation (49) calculates the robot cycle time under THTS. Constraints (58)–(60) come from Proposition 4, and we can obtain ω_{i1} and ψ_4 .

For scheduling problems, to satisfy practical application requirements, computational efficiency is a very important issue. By the above-proposed methods, a schedule is obtained by a linear program if it exists. It is well-known that a linear program can be solved in polynomial time, implying that the proposed method is efficient and scalable.

This Section analyzes the temporal properties of the four scheduling strategies (OBS, OHTS, TBS, and THTS) in multi-space process modules with RTCs. It presents the mathematical models and equations for calculating the cycle time and robot task sequences for each strategy. The analysis focuses on optimizing robot-waiting time and task scheduling to achieve minimal cycle time while meeting the RTCs. Linear programming models are developed to determine feasible and optimal schedules for maximizing the throughput. It should be noted that, with LPs 1–4, by applying different strategies, the obtained results may be different due to different parameter combinations. Thus, given a group of parameters, we need to apply LPs 1–4 and select the best schedule.

4. Experiments

Although the addressed tool is equipped with a dual-arm robot, it can be used as just a single-arm one in some cases due to the fact that the arms should be identified as dirty and clean ones. Thus, we have to present and analyze different scheduling strategies, and the performance of these strategies may be different for different parameter combinations. Essentially, given a parameter combination, we need to use all the four linear programs for a solution, then choose the best one. Nevertheless, it is better to test the strategies to see the differences in these strategies under different parameter combinations. In this way, we can provide a guide to the practitioners. To do so, based on the data from real production processes, for each test case, we randomly generate one of the parameters and keep others constant. In the following, we present parts of the experimental results to show the properties of these strategies.

We use large amounts of data collected from a fab for experiments to test the proposed method. According to the actual production situations, the processing time of wafers in different process steps is different, and the constraints on the RTC in PM are also different. At the same time, we also consider the different rotation times in PM. With these experiments, we compare the performance of the four scheduling strategies under different parameters. The time unit is seconds and omitted in the following presentation.

The algorithm is implemented using a Gurobi Python interface (gurobipy) on a personal computer equipped with Intel Core i3-10100 CPUs running at 3.60 GHz and 16.0 of RAM. To test LPs 1–4 resulting from the four scheduling strategies, a large number of experiments with different parameters are made and the results are presented as follows:

Example 1. The wafer flow pattern of this example is (1, 1, 1). Steps 1–3 are served by PM_1 , PM_2 , and PM_3 , respectively. There are three cases.

Case 1: The wafer processing time α_1 for Step 1 is randomly generated in [50.0, 100.0], $\alpha_2 = 60.0$, and $\alpha_3 = 80.0$; for RTCs, $\delta_1 = 30.0$, $\delta_2 = 40.0$, and $\delta_3 = 28.0$; and the robot task time is $\beta = 1.0$, $\beta_0 = 2.0$, and $\mu = 1.0$. The PM rotation time is $v = 2.0$. Case 1 involves variations in α_1 while keeping all other parameters constant. The results are shown in Figure 7.

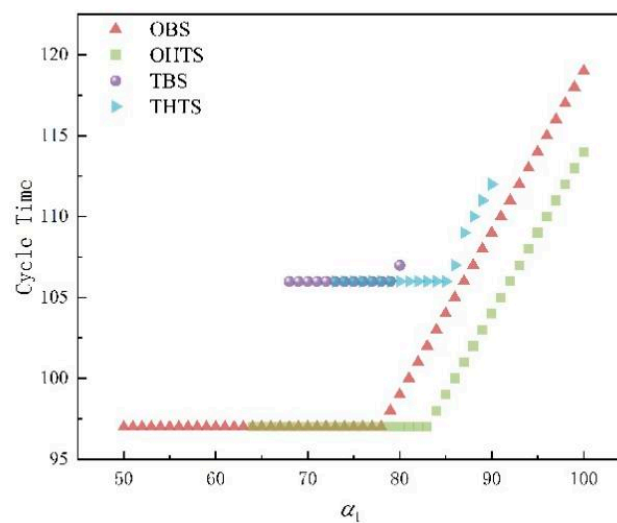


Figure 7. The cycle time varies with α_1 .

Case 2: The wafer-processing time is $\alpha_1 = 124.0$; α_2 is randomly generated in [90.0, 140.0] and $\alpha_3 = 120.0$; for RTCs, $\delta_1 = 48.0$, $\delta_2 = 45.0$, and $\delta_3 = 46.0$; the robot task times are $\beta = 2.0$, $\beta_0 = 4.0$, and $\mu = 2.0$. The PM rotation time is $v = 6.0$. Case 2 involves variations in α_2 while keeping all other parameters constant. The results are shown in Figure 8.

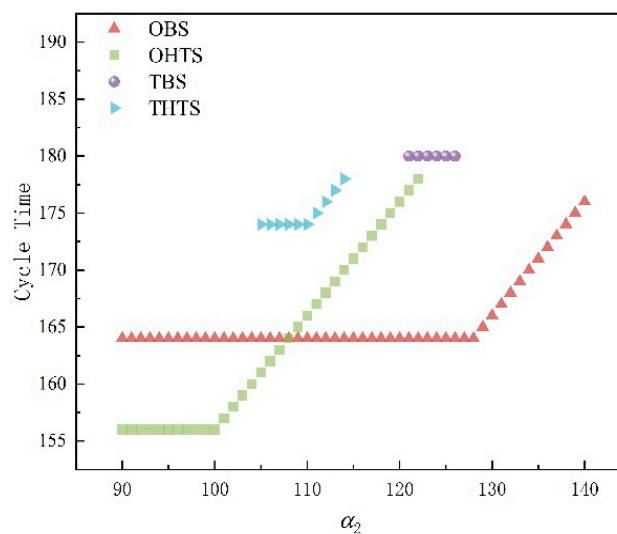


Figure 8. The cycle time varies with α_2 .

Case 3: The wafer-processing times are $\alpha_1 = 131.0$, $\alpha_2 = 135.0$, and α_3 is randomly generated in $[100.0, 160.0]$; for RTCs, $\delta_1 = 45.0$, $\delta_2 = 32.0$, and $\delta_3 = 46.0$; the robot task times are $\beta = 1.0$, $\beta_0 = 2.0$, and $\mu = 2.0$; and the PM rotation time is $v = 2.0$. Case 3 involves variations in α_3 while keeping all other parameters constant. The results are shown in Figure 9.

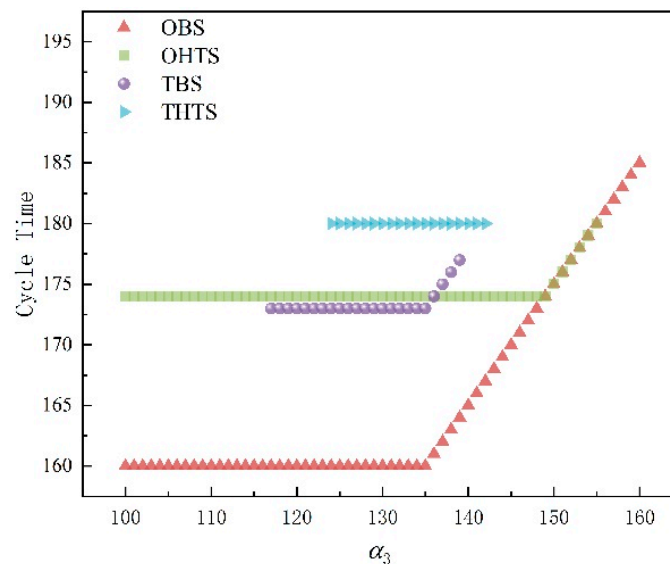


Figure 9. The cycle time varies with α_3 .

We test LPs 1–4 and calculate the parameter variation range within which feasible schedules can be found as shown in Figures 7–9 and Table 1. As can be seen from Figures 7–9 and Table 1, OBS can find schedules for the largest parameter variation range. As shown in Figures 7 and 9, when processing time for Steps 1 and 3 is short, OBS can find feasible schedules. Then, as the processing time increases, OHTS can also find feasible schedules. As shown in Figure 8, OHTS can find a feasible schedule when processing time for Step 2 is short. At the same time, the cycle time is the shortest. As the processing time increases, OBS achieves a shorter cycle time. In the given range of the parameter variation, the feasible ranges of TBS and THTS are smaller, and the cycle time is longer. In these cases, OBS and OHTS are obviously better.

Example 2. The wafer flow pattern of this example is (2, 1, 1). PM_1 and PM_2 serve for Step 1, and PM_3 and PM_4 serve for Steps 2 and 3, respectively. There are three cases.

Case 1: The wafer-processing times are $\alpha_1 = 300.0$, $\alpha_2 = 120.0$, and $\alpha_3 = 130.0$; for RTCs, δ_1 is generated randomly in $[30.0, 60.0]$, $\delta_2 = 45.0$, and $\delta_3 = 40.0$; the robot task times are $\beta = 1.0$, $\beta_0 = 2.0$, and $\mu = 1.0$; and the PM rotation time is $v = 14.0$.

Case 2: Compared to Case 1, v is reduced from 14 to 4, δ_1 is generated randomly in $[30.0, 60.0]$, with other parameters unchanged. By applying LPs 1–4, when $v = 14.0$, and $\delta_1 = 39.0, 59.0, 36.0, 51.0$, OBS, OHTS, TBS, and THTS find feasible schedules with the system cycle time $\psi_1 = 181.0$, $\psi_2 = 181.0$, $\psi_3 = 172.0$, $\psi_4 = 177.0$, respectively. As δ_1 increases, TBS finds a feasible schedule first, while OHTS finds a feasible schedule at the latest time. As for cycle time, TBS shows a shorter cycle time, followed by THTS. When $v = 4.0$ and $\delta_1 = 9.0, 13.0, 21.0, 21.0$, OBS, OHTS, TBS, and THTS find feasible schedules with the system cycle time $\psi_1 = 161.0$, $\psi_2 = 158.0$, $\psi_3 = 164.5$, and $\psi_4 = 162.0$, respectively. This exhibits that when the internal chamber of the PM rotation time is shorter, OBS performs better for a strict RTC, while OHTS has a shorter cycle, but a looser RTC.

Case 3: The wafer-processing times are $\alpha_1 = 310.0$, $\alpha_2 = 130.0$, and $\alpha_3 = 140.0$; for RTCs, $\delta_1 = 42.0$, $\delta_2 = 45.0$, and $\delta_3 = 42.0$; the robot task times are $\beta = 1.0$, $\beta_0 = 2.0$, and $\mu = 1.0$; and the PM rotation time v falls into $[2.0, 14.0]$.

By applying LPs 1–4, when the internal chamber of the PM rotation time is shorter, from Figure 10 and Table 1, we can find that OHTS performs better. With the increase in the internal chamber rotation time of a PM, TBS presents a shorter cycle time. Within the variation range of v , OBS has the widest scheduling range.

Table 1. Industrial examples.

Examples		Case 1	Case 2	Case 3	
Example 1 $WFP = (1, 1, 1)$	Parameters	$\alpha_1 = [50.0, 100.0]$, $\alpha_2 = 60.0, \alpha_3 = 80.0$; $\delta_1 = 30.0, \delta_2 = 40.0$, $\delta_3 = 28.0$; $\beta = 1.0$, $\beta_0 = 2.0, \mu = 1.0$, $v = 2.0$.	$\alpha_1 = 124.0$, $\alpha_2 = [90.0, 140.0]$, $\alpha_3 = 120.0$; $\delta_1 = 48.0, \delta_2 = 45.0$, $\delta_3 = 46.0$; $\beta = 2.0$, $\beta_0 = 4.0$, $\mu = 2.0, v = 6.0$.	$\alpha_1 = 131.0, \alpha_2 = 135.0, \alpha_3 = [100.0, 160.0]$; $\delta_1 = 45.0$, $\delta_2 = 32.0, \delta_3 = 46.0$; $\beta = 1.0, \beta_0 = 2.0, \mu = 2.0, v = 2.0$.	
	Cycle time	OBS	$\psi_1 = [97.0, 119.0]$	$\psi_1 = [159.0, 184.0]$	
		OHTS	$\psi_2 = [97.0, 114.0]$	$\psi_2 = [173.0, 180.0]$	
		TBS	$\psi_3 = [106.0, 107.0]$	$\psi_3 = [172.0, 177.0]$	
		THTS	$\psi_4 = [106.0, 112.0]$	$\psi_4 = [179.0, 180.0]$	
	Schedulable range	OBS	$\alpha_1 = [50.0, 100.0]$	$\alpha_3 = [100.0, 160.0]$	
		OHTS	$\alpha_1 = [64.0, 100.0]$	$\alpha_3 = [100.0, 156.0]$	
		TBS	$\alpha_1 = [68.0, 80.0]$	$\alpha_3 = [116.0, 140.0]$	
		THTS	$\alpha_1 = [73.0, 90.0]$	$\alpha_3 = [123.0, 143.0]$	
Example 2 $WFP = (2, 1, 1)$	Parameters	$\alpha_1 = 300.0, \alpha_2 = 120.0$, $\alpha_3 = 130.0$; $\delta_1 = [30.0, 60.0]$, $\delta_2 = 45.0, \delta_3 = 40.0$; $\beta = 1.0, \beta_0 = 2.0$, $\mu = 1.0, v = 14.0$.	$\alpha_1 = 300.0, \alpha_2 = 120.0$, $\alpha_3 = 130.0$; $\delta_1 = [30.0, 60.0]$, $\delta_2 = 45.0, \delta_3 = 40.0$; $\beta = 1.0, \beta_0 = 2.0$, $\mu = 1.0, v = 4.0$.	$\alpha_1 = 310.0, \alpha_2 = 130.0, \alpha_3 = 140.0$; $\delta_1 = 42.0, \delta_2 = 45.0$, $\delta_3 = 42.0$; $\beta = 1.0, \beta_0 = 2.0, \mu = 1.0, v = [2.0, 14.0]$.	
	Schedulable conditions	OBS	$\delta_1 = 39.0$	OBS	$\psi_1 = [164.5, 191.0]$
		OHTS	$\delta_1 = 59.0$	OHTS	$\psi_2 = [162.0, 176.0]$
		TBS	$\delta_1 = 36.0$	TBS	$\psi_3 = [168.5, 173.0]$
		THTS	$\delta_1 = 51.0$	THTS	$\psi_4 = [166.0, 177.0]$
	Cycle time	OBS	$\psi_1 = 181.0$	OBS	$v = [2.0, 14.0]$
		OHTS	$\psi_2 = 181.0$	OHTS	$v = [2.0, 9.0]$
		TBS	$\psi_3 = 172.0$	TBS	$v = [2.0, 13.0]$
		THTS	$\psi_4 = 177.0$	THTS	$v = [2.0, 9.0]$

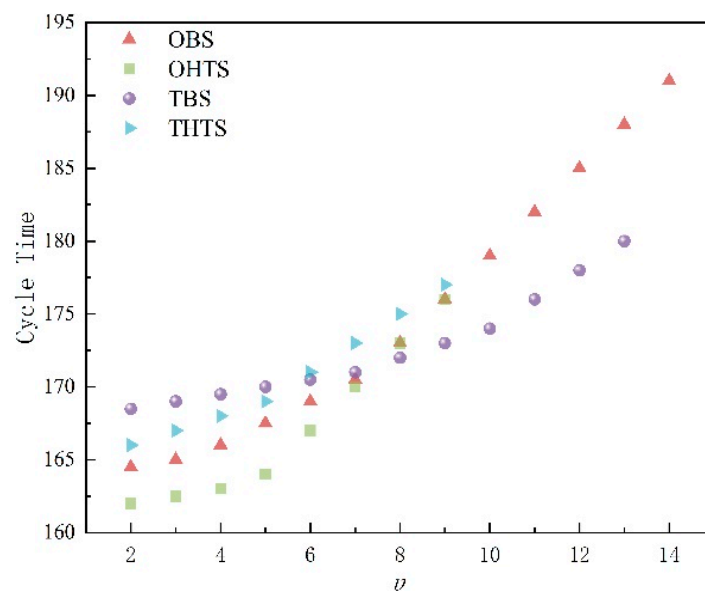


Figure 10. The cycle time varies with v .

5. Discussion

In the experiments, we test four scheduling strategies (LPs 1–4) and identify the parameter variation ranges within which feasible schedules can be found as shown in Figures 7–9 and Table 1. The experimental results demonstrate that each strategy has its strengths under specific parameter conditions. Based on the parameter variation ranges, we can recommend the best scheduling strategy for different scenarios.

OBS Strategy: As shown in Figures 7 and 9, when the processing time for Steps 1 and 3 is short, OBS can find feasible schedules. We can see in Figure 9 that when α_3 is in the range of [110, 140], OBS still achieves the shortest cycle time and the largest schedulable range. In Figure 2, we can see that when all four scheduling strategies are feasible, with $\alpha_1 = 80$ and $\psi_1 = 97$, we have that ψ_1 is 2.02% shorter than the second smallest ψ_2 (99). Within the parameter variation range, OBS covers the widest scheduling range, indicating that OBS is the best choice in cases where task complexity is low and processing time is short. Additionally, as processing time increases, OBS still achieves a relatively short cycle time. Overall, OBS is very effective in scenarios with a shorter wafer-processing time, especially when resource constraints are minimal.

OHTS Strategy: OHTS performs well in certain cases, particularly as task complexity increases. As seen in Figures 7 and 9, as processing time increases, OHTS can also find feasible schedules. When the processing time for Step 2 is short, OHTS achieves feasible schedules with the shortest cycle time (Figure 9). Therefore, OHTS is the preferred strategy when the task-processing time is longer and higher resource utilization is required.

TBS and THTS Strategies: As shown in Figure 10 and Table 1, the feasible scheduling ranges for TBS and THTS are relatively smaller, and the cycle time achieved is longer. As a result, OBS and OHTS are generally better in most cases. However, when the internal chamber rotation time of the PM increases, TBS achieves a shorter cycle time. Therefore, TBS is better in scenarios where the internal rotation time of the PM is long, while OHTS performs better when the rotation time is shorter.

From the above analysis, it is not difficult to see that the four scheduling strategies proposed in this paper also have certain limitations. When the RTC is too short, all four scheduling strategies become very difficult to find a feasible schedule due to the tool property of multi-space PMs. Similarly, when considering the RTC, if the robot's loading/unloading or movement time is too long, none of the four scheduling methods can be applied. The experimental results can provide a guide for the practitioners. Especially, when a feasible schedule cannot be found by applying all the strategies, an engineer can think about how to redesign the process such that the fabrication of a type of wafers can be completed.

It also needs to be pointed out that this study is conducted based on the specific application of wafer fabrication for the addressed cluster tools. This may be the limitation of the proposed method. Nevertheless, the idea may be useful for scheduling other cluster tools and for other similar scheduling problems.

6. Conclusions

In this paper, we examine the scheduling problem of a four-finger four-space-PM cluster tool (SF³-CT), a typical type of cluster tool with multi-space PM and a multi-finger-arm robot adopted in modern fabs. With the configuration properties, we find that both the robot task sequences and the rotation operations of the internal chambers of PMs affect the wafer stay time in PMs. Due to the increased time within PMs by dealing with multiple spaces in a PM, fulfilling the residency time constraints becomes more challenging. Consequently, finding an optimal and feasible schedule for such tools is a significant issue. To solve this problem, four robot task sequences (OBS, OHTS, TBS, and THTS) as scheduling strategies are presented, and based on them, we establish four linear programs to find optimal and feasible schedules to minimize the cycle time. Since a linear program can be solved in polynomial time, the proposed method is computationally efficient. Also, based on the property of the method, a feasible and optimal schedule can be found if it

exists. Finally, experiments are conducted with different cases to test the four strategies. It shows that for different parameter combinations, different strategies achieve different performance. Thus, given a group of parameters, one should apply all the strategies and then choose the best schedule obtained by one of the strategies.

With continuous experimentation, we have discovered that the delay time at each step is quite long. Our future endeavors will concentrate on minimizing delay time in the scheduling problem of an SF³–CT, while making efforts to reduce cycle time and enhance production efficiency.

In this paper, we focus on the scheduling strategies and solutions for finding feasible and optimal schedules for the addressed system. To do so, we assume that the activity time is known. However, in practice, uncertainty exists regarding the activity time, and this imposes another challenge for scheduling the addressed system. Today, a digital twin is a widely used technology for various application problems [33–37]. In our future work, based on the presented strategies, digital twin technology will be applied to extend the proposed method for the scheduling problem of the addressed cluster tools with activity-time uncertainty.

Currently, high-throughput screening systems are widely used for new chemical substance discovery, and their scheduling problem is also very challenging due to the residency time constraints similar to the wafer residency time constraints here [38–40]. We will examine the possibility of extending the proposed methods to study the scheduling problem of high-throughput screening systems.

Author Contributions: Conceptualization, L.G. and Y.Q.; methodology, L.G. and N.W.; software, S.Z.; validation, T.L.; formal analysis, L.G.; investigation, Y.Q.; resources, T.L.; data curation, S.Z. and T.L.; writing—original draft preparation, L.G.; writing—review and editing, N.W. and S. Z.; visualization, Y.Q.; supervision, N.W.; project administration, S.Z.; funding acquisition, N.W. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Science and Technology Development Fund (FDCT), Macau SAR (File Nos. 0018/2021/A1, 0083/2021/A2 and 0011/2023/RIA1).

Data Availability Statement: All data are contained within the article.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Bader, M.E.; Hall, R.P.; Strasser, G. Integrated processing equipment. *Solid State Technol.* **1990**, *33*, 149–154.
2. Burggraaf, P. Coping with the high cost of wafer fabs. *Semicond. Int.* **1995**, *18*, 45–50.
3. Newboe, B. Cluster tools: A process solution. *Semicond. Int.* **1990**, *13*, 82–88.
4. Singer, P. The driving forces in cluster tool development. *Semicond. Int.* **1995**, *18*, 113–118.
5. McNab, T.K. Cluster tools, part 1: Emerging processes. *Semicond. Int.* **1990**, *13*, 58–63.
6. Wu, N.Q.; Chu, C.B.; Chu, F.; Zhou, M.C. A Petri net method for schedulability and scheduling problems in single-arm cluster tools with wafer residency time constraints. *IEEE Trans. Semicond. Manuf.* **2008**, *21*, 224–237. [[CrossRef](#)]
7. Perkinson, T.L.; Gyurcsik, R.S.; MacLarty, P.K. Single-wafer cluster tool performance: An analysis of the effects of redundant chambers and revisitation sequences on throughput. *IEEE Trans. Semicond. Manuf.* **1996**, *9*, 384–400. [[CrossRef](#)]
8. Rostami, S.; Hamidzadeh, B.; Camporese, D. An optimal periodic scheduler for dual-arm robots in cluster tools with residency constraints. *IEEE Trans. Robot. Autom.* **2001**, *17*, 609–618. [[CrossRef](#)]
9. Lee, T.-E.; Park, S.-H. An extended event graph with negative places and tokens for time window constraints. *IEEE Trans. Autom. Sci. Eng.* **2005**, *2*, 319–332. [[CrossRef](#)]
10. Lopez, M.-J.; Wood, S.-C. Systems of multiple cluster tools: Configuration, reliability, and performance. *IEEE Trans. Semicond. Manuf.* **2003**, *16*, 170–178. [[CrossRef](#)]
11. Lee, T.-E.; Lee, H.-Y.; Shin, Y.-H. Workload balancing and scheduling of a single-armed cluster tool. In Proceedings of the 5th APIEMS Conference, Gold Coast, QLD, Australia, 12–15 December 2004; pp. 1–6.
12. Venkatesh, S.; Davenport, R.; Foxhoven, P.; Nulman, J. A steady-state throughput analysis of cluster tools: Dual-blade versus single-blade robots. *IEEE Trans. Semicond. Manuf.* **1997**, *10*, 418–424. [[CrossRef](#)]
13. Wang, J.; Hu, H.; Pan, C.; Zhou, Y.; Li, L. Scheduling dual-arm cluster tools with multiple wafer types and residency time constraints. *IEEE/CAA J. Autom. Sin.* **2020**, *7*, 776–789. [[CrossRef](#)]

14. Yi, J.; Ding, S.; Song, D.; Zhang, M.T. Steady-state throughput and scheduling analysis of multi-cluster tools: A decomposition approach. *IEEE Trans. Autom. Sci. Eng.* **2008**, *5*, 321–336.
15. Ding, S.; Yi, J.; Zhang, M.T. Multicenter tools scheduling: An integrated event graph and network model approach. *IEEE Trans. Semicond. Manuf.* **2006**, *19*, 339–351. [[CrossRef](#)]
16. Perkinson, T.L.; McLarty, P.K.; Gyurcsik, R.S.; Cavin, R.K. Single-wafer cluster tool performance: An analysis of throughput. *IEEE Trans. Semicond. Manuf.* **1994**, *7*, 369–373. [[CrossRef](#)]
17. Chan, W.K.V.; Yi, J.; Ding, S. Optimal scheduling of multicenter tools with constant robot moving times, part I: Two-cluster analysis. *IEEE Trans. Autom. Sci. Eng.* **2011**, *8*, 5–16. [[CrossRef](#)]
18. Zuberek, W.M. Timed Petri nets in modeling and analysis of cluster tools. *IEEE Trans. Robot. Autom.* **2001**, *17*, 562–575. [[CrossRef](#)]
19. Kim, J.-H.; Lee, T.-E.; Lee, H.-Y.; Park, D.-B. Scheduling analysis of time-constrained dual-armed cluster tools. *IEEE Trans. Semicond. Manuf.* **2003**, *16*, 521–534. [[CrossRef](#)]
20. Lee, J.-H.; Kim, H.-J.; Lee, T.-E. Scheduling lot switching operations for cluster tool. *IEEE Trans. Semicond. Manuf.* **2013**, *26*, 592–601. [[CrossRef](#)]
21. Wu, N.Q.; Zhou, M.C. A closed-form solution for schedulability and optimal scheduling of dual-arm cluster tools with wafer residency time constraint based on steady schedule analysis. *IEEE Trans. Autom. Sci. Eng.* **2010**, *7*, 303–315.
22. Lim, Y.; Yu, T.-S.; Lee, T.-E. A new class of sequences without interferences for cluster tools with tight wafer delay constraints. *IEEE Trans. Autom. Sci. Eng.* **2019**, *16*, 392–405. [[CrossRef](#)]
23. Ko, S.-G.; Yu, T.-S.; Lee, T.-E. Wafer delay analysis and workload balancing of parallel chambers for dual-armed cluster tools with multiple wafer types. *IEEE Trans. Autom. Sci. Eng.* **2021**, *18*, 1516–1526. [[CrossRef](#)]
24. Lim, Y.; Yu, T.-S.; Lee, T.-E. Adaptive scheduling of cluster tools with wafer delay constraints and process time variation. *IEEE Trans. Autom. Sci. Eng.* **2020**, *17*, 375–388. [[CrossRef](#)]
25. Zhu, Q.H.; Wang, G.H.; Hou, Y.; Qiao, Y.; Wu, N.Q.; Zhou, M.C. Optimally scheduling dual-arm multi-cluster tools to process two wafer types. *IEEE Trans. Robot. Autom.* **2022**, *7*, 5920–5927. [[CrossRef](#)]
26. Yang, F.J.; Wu, N.Q.; Qiao, Y.; Zhou, M.C.; Li, Z.W. Scheduling of single-arm cluster tools for an atomic layer deposition process with residency time constraints. *IEEE Trans. Syst. Man Cybern. Syst.* **2017**, *47*, 502–516. [[CrossRef](#)]
27. Qiao, Y.; Wu, N.Q.; Zhou, M.C. Schedulability and scheduling analysis of dual-arm cluster tools with wafer revisiting and residency time constraints based on a novel schedule. *IEEE Trans. Syst. Man Cybern. Syst.* **2015**, *45*, 472–484. [[CrossRef](#)]
28. Qiao, Y.; Wu, N.Q.; Zhou, M.C. Petri net modeling and wafer sojourn time analysis of single-arm cluster tools with residency time constraints and activity time variation. *IEEE Trans. Semicond. Manuf.* **2012**, *25*, 432–446. [[CrossRef](#)]
29. Wu, N.Q.; Zhou, M.C. Modeling, analysis and control of dual-arm cluster tools with residency time constraint and activity time variation based on Petri nets. *IEEE Trans. Autom. Sci. Eng.* **2012**, *9*, 446–454.
30. Yu, T.-S.; Kim, H.-J.; Lee, T.-E. Scheduling single-armed cluster tools with chamber cleaning operations. *IEEE Trans. Autom. Sci. Eng.* **2018**, *15*, 705–716. [[CrossRef](#)]
31. Yu, T.-S.; Lee, T.-E. Scheduling dual-armed cluster tools with chamber cleaning operations. *IEEE Trans. Autom. Sci. Eng.* **2019**, *16*, 218–228. [[CrossRef](#)]
32. Yu, T.-S.; Lee, T.-E. Wafer delay analysis and control of dual-armed cluster tools with chamber cleaning operations. *Int. J. Prod. Res.* **2019**, *58*, 434–447. [[CrossRef](#)]
33. Pires, F.; Cachada, A.; Barbosa, J.; Moreira, A.P.; Leitao, P. Digital twin in Industry 4.0 : Technologies, applications and challenges. In Proceedings of the 2019 IEEE 17th International Conference on Industrial Informatics (INDIN), Helsinki, Finland, 22–25 July 2019; pp. 721–726.
34. Borys, S.; Kaczmarek, W.; Laskowski, D. Selection and optimization of the parameters of the robotized packaging process of one type of product. *Sensors* **2020**, *20*, 5378. [[CrossRef](#)]
35. Ullrich, M.; Thalappully, R.; Heieck, F.; Lüdemann-Ravit, B. Virtual commissioning of linked cells using digital models in an industrial metaverse. *Automation* **2024**, *5*, 1–12. [[CrossRef](#)]
36. Kaczmarek, W.; Lotys, B.; Borys, S.; Laskowski, D.; Lubkowski, P. Controlling an industrial robot using a graphic tablet in offline and online mode. *Sensors* **2021**, *21*, 2439. [[CrossRef](#)]
37. Ugarte, M.; Etxeberria, L.; Unamuno, G.; Bellanco, J.L.; Ugalde, E. Implementation of digital twin-based virtual commissioning in machine tool manufacturing. *Procedia Comput. Sci.* **2022**, *200*, 527–536. [[CrossRef](#)]
38. Mayer, E.; Raisch, J. Time-optimal scheduling for high throughput screening processes using cyclic discrete event models. *Math. Comput. Simul.* **2004**, *66*, 181–191. [[CrossRef](#)]
39. Mayer, E.; Haus, U.-U.; Raisch, J.; Weismantel, R. Throughput-optimal sequences for cyclically operated plants. *Discret. Event Dyn. Syst.* **2008**, *18*, 355–383. [[CrossRef](#)]
40. Wu, N.Q.; Qiao, Y.; Li, Z.W.; Al-Ahmari, A.; El-Tamimi, A.; Kaid, H. A novel control-theory-based approach to scheduling of high throughput screening system for enzymatic assay. *IEEE Trans. Syst. Man Cybern. Syst.* **2022**, *52*, 7667–7678. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.